

EXPERIMENTAL INVESTIGATION AND MANUFACTURING OF CLAY BLOCKS USING RED MUD, RICE HUSK AND WATER HYACINTH

PROJECT REPORT

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to

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of

Bachelor of Technology

in

Civil Engineering



Department of Civil Engineering

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DECLARATION

We undersigned hereby declare that the project report “Experimental Investigation and Manufacturing of Clay Blocks using Red Mud, Rice Husk and Water Hyacinth”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Ms. Midhila V S, Asst. Prof., Department of Civil Engineering. This submission represents our ideas in our own words and where ideas or words of others have been included; we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and / or the University and can also evoke penal action from the sources which have thus not been obtained. This report has not been previously formed the basis for the award of any degree, diploma, Por similar title of any other University.

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This is to certify that the report entitled '**Experimental Investigation and Manufacturing of Clay Blocks using Red Mud, Rice Husk and Water Hyacinth**' submitted by '**Amaljith S, Anandhu Krishnan M, Arya Aji, and Pramath K S**' to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Civil Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Tiles and clay blocks are a conventional part of the construction industry. From being used in the roofs, floors, and walls all forms of blocks upgrade the aesthetics of the space, be in interior or exterior. Moreover, these blocks when used on floor protect against endure loads and abrasions well enough. In the present scenario of finding sustainable alternatives, innovations in blocks manufacturing are profound. Additions of waste materials into the synthesis of blocks are ingenious and worthwhile in the reduction of usage of natural resources. Proper combinations and unifications of the waste into the well-known utilities of the human life can prove to be very beneficial.

Water hyacinth is an invasive weed depreciating the water quality, whereas red mud is the by-product waste from the Bayer's aluminium production process and rice husk is a coating on seed of rice of grain. Effective utilization of these into the manufacturing of blocks can be a salutary approach. This project aims to fabricate clay blocks using various consolidation of water hyacinth, red mud, rice husk and a combination of the three into the conventional blocks formulation. The raw materials used for the manufacturing of clays blocks were collected and tested for Standard proctor compaction test, grain size distribution analysis, Atterberg limit and which on satisfy the acceptable values from Indian standard codes. Bricks composed in part by red mud would reduce the weight of structures, ultimately reducing building and foundation costs, as it helps to improve flexural strength, durability to brick and also helps to reduce the shrinkage of adobe brick. A case study approach from the existing works is adopted for the procedure of the project.

Keywords: Red mud, Rice husk, Water hyacinth, Waste, Recycling, Clay blocks

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ABBREVIATION

GHG	Green House Gas
pH	Potential of Hydrogen
PL	Plastic Limit
RHA	Rice Husk Ash
SCM	Supplementary Cementitious Material
SI	Shrinkage index
SW	Solid Waste
RM	Red Mud
WH	Water Hyacinth

NOTATIONS

g	Gram
kg	Kilogram
M	Mass, (g or kg)
V	Volume, (mm)
wt. %	Percentage weight
<i>Chemical formula</i>	
CO ₂	Carbon dioxide
NO	Nitric oxide
PM ₁₀	Particulate matter (less than 2.5 micrometre dia.)
SO	Sulphur monoxide
<i>Greek symbols</i>	
ρ	Density
μ	Microns (1×10 ⁻⁶)
%	Percentage
<i>Subscripts</i>	
C _C	Coefficient of curvature
C _U	Uniformity coefficient
D ₁₀	Particle size at 10% finer
D ₃₀	Particle size at 30% finer
D ₆₀	Particle size at 60% finer
<i>Superscripts</i>	
°C	Degree Celsius

CHAPTER 1

INTRODUCTION

1.1 GENERAL BACKGROUND

Environmental contamination due to solid waste mismanagement is a global issue. Open dumping and open burning are the main implemented waste treatment and final disposal systems, mainly visible in low-income countries. General increase in accumulation of unmanaged agriculture and industrial waste in the world has been a major concern. Agricultural-based industries produced the vast number of residues every year. If these residues are released to the environment without proper disposal procedure that may cause to environmental pollution and harmful effect on human and animal health. Most of the agro-industrial wastes are untreated and underutilized. These untreated wastes create different problems with climate change by increasing several greenhouse gases. Besides this, the use of fossil fuels also contributing the effect on greenhouse gases (GHG) emission. Uncontrolled disposal generates serious heavy metals pollution occurring in the water, soil, and plants, open burning is cause of CO, CO₂, SO, NO, PM₁₀ and other pollutant emissions that affect the atmosphere, waste picking within open dump sites pose to serious health risk people working on these areas, release of SW in water bodies improve the marine litter globally, enhancing environmental contamination. Therefore, SW mismanagement is cause of sever and various environmental and social impacts, which do not allow improvements in sustainable development. An effective way to deal with this increasing waste is to use this waste for sustainable construction of roads, buildings etc.

The main aim of research in present days is to enhance environmental effectiveness and to implement a comprehensive construction management plan. More precisely this study presents how can we reuse the increasing waste materials especially water hyacinth, rice husk and red mud in a good way for creating sustainable construction structures and reduce the amount of waste decomposition which will lead to eco-friendly environment. The demand and cost of construction materials is increasing day by day, as a result there is scarcity of raw materials hence, to save the natural resources having an alternative construction material is a global concern. Protection of nature has become a major concern from the environmental point of view. Construction field is a prominent field in handling this waste to reuse them as

construction materials. Researchers are working on producing prominent and effective construction materials from waste produced.

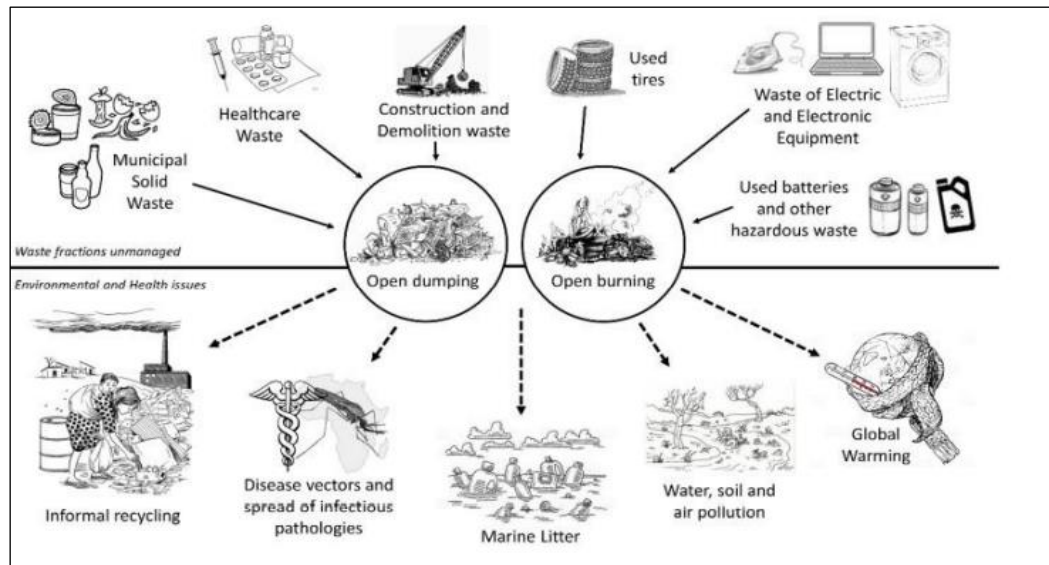


Fig.1.1 Problem related to waste (www.google.com)

There is a tremendous scope for setting up secondary industries for recycling and using such huge quantity of solid wastes as minerals or resources in the production of construction materials. Environment-friendly, energy-efficient, and cost-effective alternative materials produced from solid wastes will show a good market potential to fulfil people's needs in rural and urban areas. The construction products from solid wastes practically will not be useful despite their good mechanical properties if the durability performance is not satisfactory. Therefore, rigorous quality control and assurance should be practiced for durability improvement while using solid wastes in construction materials. In addition, the impact of solid wastes on the durability performance of construction materials must be properly assessed before commercialization. Solid wastes can be used as coarse aggregates, fine aggregates, or supplementary cementing material for the production of construction materials. The raw materials used are water hyacinth, red mud, rice husk, clay and water.

In this project we use red mud, a byproduct of the alumina manufacturing process, can be utilized in place of clay in some block-making processes. This may lower the production costs. Other agricultural wastes, such as rice husk and water hyacinth, both of which are regarded as waste items, can also partially replace clay. We will be casting blocks in this research phase utilizing a number of mix designs, each of which will have a different proportion of clay, red mud, rice husk, and water hyacinth.

The blocks are then put through compression tests to determine their strength and tests to determine how well they absorb water to determine how long they will endure.

1.2 CLAY BLOCKS

Clays and clay minerals are very important industrial minerals. There are several documented industrial applications of clay materials. The physical and chemical properties of the clay minerals determine its use as an industrial material because applications of the clay minerals depend on their compositions, structures and physical properties. Burnt clay blocks products are produced from natural materials containing a high proportion of clay minerals. Due to the development, clay deposits have an economic importance. Since clay blocks are durable, as they are built to last. Because they can withstand a lot of wear and tear. These blocks are also resistant to fire and water, which also makes it a great choice. Apart from this they give a good aesthetic appearance.



Fig. 1.2 Clay blocks (www.blogedu.com)

1.3 NEED AND SCOPE

As, said in introduction the primary aim of the project is to enhance environmental effectiveness and to implement a new method of construction management. Also, it shows how the increasing in the reusing of waste materials especially water hyacinth with create harmful effect on nature. The demand and cost of construction materials is increasing day by day, as a result there is scarcity of raw materials hence, to save the natural resources having an alternative construction material.

A new way to produce the clay block is proposed by adding 3 main waste materials which include red mud, rice husk and water hyacinth. Aims to increase the utilization of waste materials into the construction. Safe and innovative use of the waste

and reduction in the natural resource's usage. Production of economically safe and sound tiles for utilization.

1.4 OBJECTIVES

- To manufacture blocks with 100% clay composition and determine its strength and other parameters.
- To perform the qualitative analysis in each specimen in varying proportions.
- To study the strength and properties of varying percentage for each combination.
- To identify the best mix ratio from each specimen.
- To determine the property of specimen in combination of Clay, Red mud, Rice husk and Water hyacinth.
- To evaluate the ideal ratio from different proportion against typical clay block for comparison.

1.5 ORGANIZATION OF THE PROJECT

The works carried out during the investigation are presented systematically in different chapters as follows:

1. Chapter 1 comprises the introduction, objectives, and need and scope of the project.
2. Chapter 2 comprises the literature review for the project which consists of studies and results on soils.
3. Chapter 3 comprises the methodology of the project.
4. Chapter 4 comprises the study of clay blocks, material properties and test results.
5. Chapter 5 comprises the study of clay blocks partially replaced by rice husk, its property testing and analysis of results.
6. Chapter 6 comprises the study of clay blocks partially replaced by red mud, its property testing and analysis of results.
7. Chapter 7 comprises the study of clay blocks partially replaced by water hyacinth, its property testing and analysis of results.
8. Chapter 8 comprises the study of clay blocks with red mud, rice husk and water hyacinth, its testing and comparison with standard clay blocks.
9. Chapter 9 comprises the conclusion.

CHAPTER 2

LITERATURE REVIEW

2.1 GENERAL

A literature review is an overview of the previously published works on a specified topic. This chapter includes the literature survey related to the project work which give an overview of procedures and evaluation on different soil as well as material testing that can be done to determine the properties of samples and the varying combinations also.

2.2 EXISTING LITERATURE

Belfiore CM et. al (2022)^[1] investigated the possible re-use of these volcanic ashes as a plastic component in the manufacture of ceramic tiles. Analytics methods done were characterization of volcanic materials and characterization of ceramic tiles. Determination of water absorption, determination of resistance to bending, determination of impact resistance by measurement of the coefficient of restitution, determination of resistance to deep abrasion, determination of resistance to thermal shock, determination of frost resistance, accelerated aging test by salt crystallization. Results obtained proved to be absolutely encouraging, so providing a stimulus in the continuation of the research.

Zanatta T et.al (2021)^[13] performed test on ceramic tiles materials and compared it to the commercial ones. Chamotte (CT), sludges from the treatment of residual water ceramic sludge (CS), and raw wastes (RW) were mixed into one formulation for each pure component, three formulations for binary mixtures with equal proportions of components and a better composition with equal proportions of components and tiles were fabricated. The fabrication was then sintered in a laboratory kiln at 1120, 1160, and 1200 °C. It was tested for water absorption, breaking strength, linear shrinkage, bulk density and modulus of rupture. With this the tiles were categorized into wall, floor and porcelain stoneware tiles.

Moussi B and Hajjaji W (2020)^[4] analysed the Numidian clay from 2 different region and the usage of the clay in ceramic tile manufacturing is investigated. The clay is collected, and the characteristics of the clay interpreted. The chemical composition of the clay is found out by XRF while the mineralogical analyses of bulk samples were

carried XRD. The TGA, grain size, DTA were all performed on the sample clay. The clay was then formulated either complexly using one variant or by a mixture of the clay variant with feldspar and quartz. The tiles were fabricated and tested for its mannerism in the water absorption, flexural strength, and firing shrinkage under 1000, 1050, 1100, 1500 °C. The TDA and SEM of the tiles casted were taken and the results suggested that those variants of the clay with mixture formulation could be used for the fabrication as the results were in appreciable range. The flexural strength and water absorption of the variants were in credited limits and the shrinkage value was comparably less.

Zanelli et.al (2020)^[15] studied the effects of waste recycling on technological behaviour and technical performance of ceramic tiles, with some concern on the environmental impact. However, any life cycle assessment of waste addition in tile-making is not considered here. Technical constraints that waste recycling must comply to be transferable to ceramic tile manufacturing were examined in detail and available information on the behaviour of ceramic tile bodies containing wastes from various sources was critically reviewed for every manufacturing stage. The conclusion they found was that the main obstacle to a widespread recourse to waste recycling is the lack of knowledge about the effect on technological behaviour and technical performance of ceramic tiles, these aspects have been examined in detail, in order to outlook feasibility, recyclable amount, and possible matters of concern.

Hossain ST8S and Ranjan V (2020)^[3] studied waste materials such as blast furnace slag, electric arc furnace slag, steel plant dust, paper mill sludge, petroleum waste, rice husk ash, rice straw ash, sugarcane bagasse ash, coffee husk ashes, fishbone ash, fly ash, ceramic waste and glass waste for tiles production. This study has investigated the feasibility of using river silt as an alternative raw material for the manufacturing of ceramic tiles by replacing quartz, studies the effect of waste derived wollastonite addition in place of feldspar on the physical and mechanical performance of tiles with different sintering temperatures. River silts, quartz, ball clay, K-feldspar and wollastonite as raw materials. The physical and mechanical properties of the tile samples are reduced with the incorporation of river silt in the place of quartz for low temperature sintering. The colour of the sample is changing from light to darkness with silt addition.

Sukarni S et.al (2019)^[9] examined thoroughly the potential of WH as a fuel which includes its chemical properties, physical properties (moisture content, volatile content, fixed carbon, ash), and calorific value. Tests involved are chemical composition test (tested using Energy-dispersive X-ray (EDX) spectrometry), and Proximate and heating value test (to determine its content of moisture, volatile matter, fixed carbon, and ash). The test results shows that biomass is predominantly with O element. WH may possibly become the coal mixture to increase coal's reactivity during the combustion process.

Xu X et.al (2019)^[12] experimented the utilisation of red mud in the ceramic industry works. Four different red mud (A,B,C,D) were collected from different areas and mixed at 40% wt.% with quartz, talc, flint lay, and shale. The raw materials were tested, grounded, wet mixed and prepared into the tiles. The 4 samples had different sintering temperatures and it was tested for water absorption, density, porosity, 3-point bending etc and the red mud was tested for its radioactivity levels. The tested conducted on the specimens showed that the 1st formulation of tiles had better properties even at the lower temperature thereby making it more energy efficient.

Zanelli C et.al (2019)^[14] experimented the addition of boron residue muds into the ceramic tile production and the analysis were done. the boron mud was added into the formulation with 2-5-10% wt.%. 2 fabrications we were done by one relacing ball clay and another replacing the a feldspar. Chemical analysis, mineralogical composition, test fusibility of the raw materials was tested, and the results calculated. The results showed that the firing shrinkage increased for one formula while the other was comparatively stable the water absorption and densification of both the sample was pretty appreciable. The addition of 5% wt.% sounded apt above which there is a hinderance in the production.

Njindam ZR et.al (2018)^[5] studied the inclusion of waste glass powder with clay mixture for the manufacturing of porcelain stoneware wall and floor tiles. The clay materials used are Kaolinite clay and plastic clay, while the glass raw material is obtained from glass waste of green and colour less bottles. Mixtures of different proportions are made, and the tests are carried out. It was found out that linear shrinkage increases with glass powder content up to 30 wt.%. The values of water absorption of tiles decrease with the increase in waste glass and firing temperature. Thus, fired products containing 30 wt.% of glass showed the weakest water absorption value

(0.4%). Bulk density also increases with increase in waste glass addition and firing temperatures. The study also revealed that, the flexural strength of the ceramic products increases and reaches a maximum. Furthermore, the formation of some crystalline phases like cristobalite low, anorthite, enstatite and mullite is an advantage for sintered glass composite as it exhibits excellent mechanical properties.

Rodrigues LP et.al (2018)^[6] performed the research to find the adaptability of using municipal waterworks sludge into the ceramic tiles production industry. The municipal waste was added into the ceramic formulation as four floor tile formulations containing up to 10 wt. % to replace the kaolin. The floor tile processing route consisted of dry powder granulation, uniaxial pressing, and firing between 1190 and 1250 °C. A characteristic study of the raw materials was initially done and the tabulated the results. The tested conducted on the tiles specimens showed that the waster absorption of the specimens decreased while the linear shrinkage increased but the tiles ad good dimensional control. The fast-firing sequence led to the densification of the tiles and resulted in advanced densities of the tiles.

Scribot C et.al (2018)^[7] conducted a study on the addition of red mud into the ceramic material production. Here the red mud was added into the clay obtained in the ratios of 0-30% wt.%. the formulations were grounded and powdered and made into tiles. It was sintered under 950 and 1015 °C. The density, porosity, elementary chemical composition, crystalline phases, environmental impacts of the raw materials were calculated initially. The fabricated tiles were then subjected to linear shrinkage, water absorption, flexural and unconfined compressive strength, bulk density etc. The results showed that at a greater temperature of 1015 °C the 30% addition formulae showed better compressive and flexural strength while the bulk densities remained comparatively the same as compared to the commercial tiles.

Siqueira FB and Holanda JN (2018)^[8] focused on the reuse of grits waste, from cellulose industry, as a raw material to replace traditional carbonate material in ceramic wall tiles. Wall tile compositions were formulated using mixtures of kaolin, quartz, calcareous, and grits waste. The tests and experiments are undergone. The experimental data showed that the grits waste can totally replace traditional calcareous material in ceramic wall tile for mutations. The grits waste positively influenced the densification

behaviour, physical and mechanical properties and fired microstructure of the wall tile specimens.

Wang W et.al (2018)^[11] studied on the originality of the strengthening of ceramic floor tile with low density by the synthetic rod-like mullite from red mud and Kaolin, with molybdenum oxide as crystallization catalyst. The raw materials used are red mud, kaolin and analytically pure ammonium molybdate. The high strength lightweight mullite-ceramic can be used for producing ceramic floor tile. This method also supplies an effective way to recycle waste red mud in industries. The alkali metal oxides in red mud can be used as sintering aids to lower sintering temperature for every waste source.

De Silva GS and Surangi ML (2017)^[2] studied on the effect of waste rice husk ash on properties of clay roof tiles. Clay and rice husk are the basic raw materials. The raw materials are mixed and the tile manufacturing process is undergone. Several tests are done and it was found out that 10% wt. % replacement of clay with RHA shows 45.97% increment in the transverse breaking load compared to that of the conventional clay roof tile along with better compressive strength, reduced linear shrinkage, ductility. The densification the clay reduces with RHA addition indicating that the replacement will lead to light-weight roof tiles production scope.

Sultana MS et.al (2015)^[10] researched about the roof tiles which are manufactured with maximum utilization of rock dust sintering at low temperatures. Tiles were prepared with different percentages of rock dust (10- 50%) mixed with clay sintered from 850-1100 °C. The fired specimens were subjected to various physical analyses like Bulk density, water absorption and apparent porosity by Archimedes' immersion technique on keeping the sample in boiling water followed by ASTM C20-00. Plasticity of clay sample by Atterberg's method. Linear shrinkage was determined by the test sample length after drying at 110 °C and the sample length after firing at different temperatures. The mechanical strength and Young's modulus were studied by UTM. Particle size of clay was determined with laser scattering particle size analyser. The microstructure of tiles body was examined by SEM. Rock dust has been found good for using as fluxing material after XRF study (X-ray fluorescence Spectrometer). After the tests it is concluded that the behaviour of water absorption as firing temperature was better for 20% and 30% rock dust than other samples. Mechanical strength of the samples is within the range of 31–53 MPa. Shrinkage values were low thus it reduces

the risk of defects such as warping and cracking. Also, the possibility of producing roof tiles incorporating up to 40% of rock dust having better properties fired at 900 °C.

2.3 CONCLUSION FROM LITERATURE REVIEW

The literature review helps in the methodology and analysis of the project. Red mud has high water retention capacity due to the presence of clay minerals and their particle fineness. Higher fabricated temperatures produce better strength in the tiles. Water hyacinth contains 92-95% water and proper drying of the material is required. Exact blocks formulation is insufficient for a proper comparison. Due to the regional availability and differences, the chemical and physical characters of the waste materials may differ and cause changes in fabrication. The results interpretation is highly process oriented. Addition of waste materials in the fabrication without causing intolerance needs to be tested before every waste material addition for proper results. Apart from that there is a final combination consisting of the materials like clay, red mud, rice husk and water hyacinth altogether.

CHAPTER 3

METHODOLOGY

3.1 GENERAL

The methodology consists of selection of raw materials and there by analysing the property of each constituent.

3.2 STEPS INVOLVED

The step for this experiment is represented by using flow chart in given below:

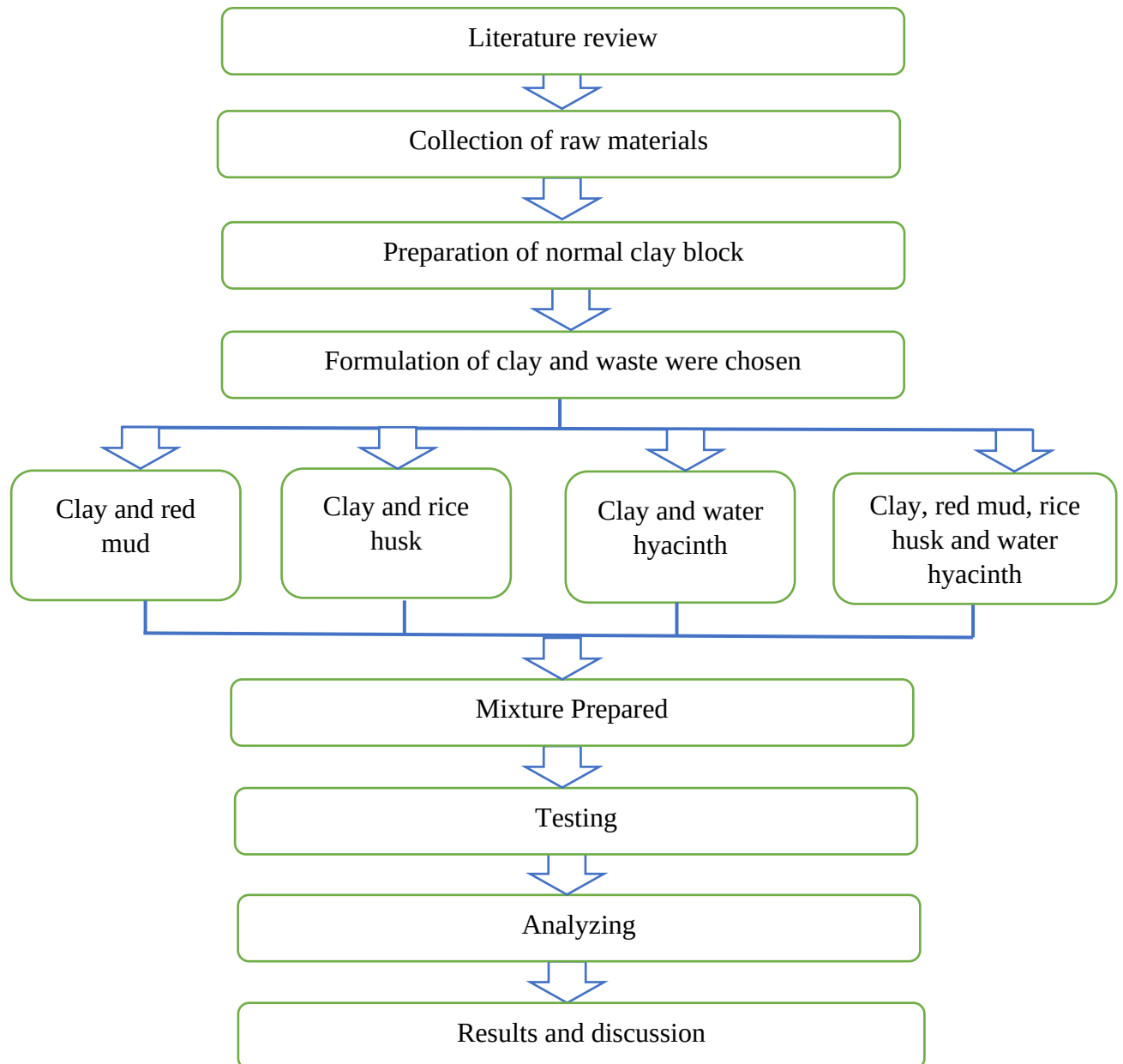


Fig. 3.1 Flowchart of methodology

3.3 MATERIALS USED

The main raw materials used as red mud, rice husk and water hyacinth.

3.3.1 Clay

Clay is a form of fine-grained soil that is mostly made up of mineral particles such as hydrated silicates and aluminum silicates. Clay is also known as "kaolin." It finds widespread application in a wide variety of fields, including ceramics, architecture, and the arts. It has a property known as plasticity, which enables it to be moulded and formed when it is wet and then to retain its shape when it is dry. Because of its quality, it can be utilized in ceramics as well as other forms of art. Its cohesive quality comes from the fact that the mineral particles within it are held together by electrostatic forces, which also contribute to clay's cohesiveness. Because of its quality, it can be beneficial in the construction industry, more specifically in the production of clay constructions such as adobe buildings. The average diameter of a clay particle is less than 0.002 millimeters, making it a fine-grained material.



Fig.3.2 Clay (www.google.com)

Clay can have a variety of colours, although most commonly it is of red, brown, or grey. The colour of clay can change depending on the type of clay and the content of the clay, though. They can have a variety of textures, ranging from smooth to gritty, depending on the type of clay and the components that make up the clay. It is a versatile material that can be used for a broad variety of purposes due to the physical features that it possesses. Some examples of these applications are pottery and art.

3.3.2 Red mud

Red mud is a by-product produced in the process of extraction of alumina from bauxite. The process is called Bayer's Process. Its insoluble product and is generated after bauxite digestion with sodium hydroxide at elevated temperature and pressure. Depending on the raw material processed, 1–2.5 tons of red mud is generated per ton of alumina produced. In India, about 4.71 million tons/annum of red mud is produced which is 6.25% of world's total digestion with sodium hydroxide at elevated temperature and pressure. It is a mixture of compounds originally present in the parent mineral bauxite and of compounds formed or introduced during the Bayer cycle. It is disposed as slurry having a solid concentration in the range of 10-30%, pH in the range of 10-13 and high ionic strength. Red mud contains much aluminium, iron and contains rare metals such as scandium, titanium, and vanadium. The red mud produced from different processes has big difference in respect of appearance and microstructure, so that different red muds have their physical and mechanical properties.

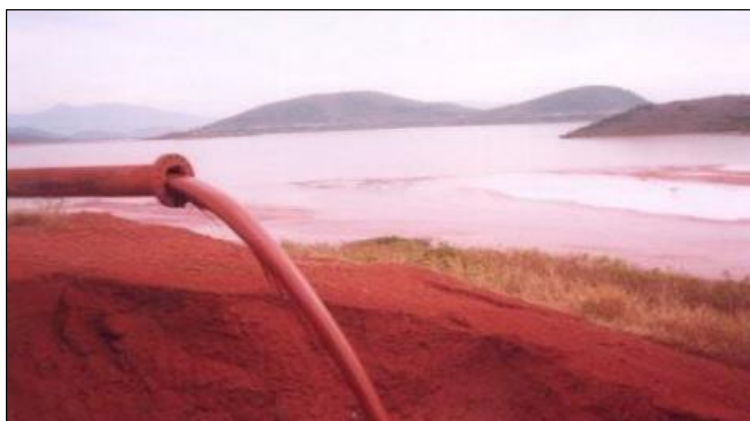


Fig. 3.3 Discharge of red mud into red mud pond
(www.google.com)

The environmental impacts caused by red mud are many. Ground water pollution-when the red mud gets mix with water. Alkali seepage into underground water resources such as wells, aquifer may get polluted. Alkaline air born dust fly with air and effects on transpiration process of plant which result in reduction of plant life. Land disposal changes the property of soil and result in lesser fertility.

Table.3.1 Composition percentage of red mud

Fe_2O_3	30-60%
Al_2O_3	10-20%
SiO_2	3-50%
Na_2O	2-10%
CaO	2-8%
TiO_2	25%

Finishing is the final process of construction of a building. Choosing the appropriate blocks material is an important activity in the construction of a building. Finding out more cost effective and environment friendly materials have become topics of immense importance. Our study aims at finding out an appropriate combination of water hyacinth, red mud and clay which together results in high quality bricks. Thereby doing our bit for the conservation of the environment.

3.3.3 Rice husk

The rice husk is the outermost part of the rice grain which is a by-product when the rice milling process is done. About 20-35 percent of the weight of rice is rice husk and approximately fifteen percent of the rice husk composition is husk ash. Silica dioxide (amorphous) from rice husks through several stages of the process, i.e., washing, drying, spraying, bleaching, and acidification. The highest content of SiO_2 obtained by drying with sunlight for 1 hour that is equal to 88 - 97%, compared with drying in oven (190 °C) 1 hour which equal to 83.15%. It has a high silica content, with around 20% of its total composition being made up of the mineral and has a low density, having a density of roughly 96 kg/m³, making it a lightweight material. Rice husk has a porous structure, which enables it to absorb water and other liquids. This property makes it useful in agricultural applications.

It is a great material for applications involving adsorption and filtration because it has a high surface area and a porous structure. Rice husk's high surface area is owing to its porous nature. Because it has a low thermal conductivity, rice husk is an excellent insulator. Rice husk has a low thermal conductivity. It is also a potent source of pozzolan as SCM (Supplementary Cementitious Material). RH has a very high pozzolanic content that is superior to other SCM like fly ash, slag, and silica fume.



Fig.3.4 Rice husk (www.google.com)

The silica content in RH can be utilized in the construction world as an added material to increase the compressive strength. The added material in the concrete used today is the silica fume produced by the factory, so the price is very expensive. Another case with RH which has a silica content which is the waste material from burning rice husks much cheaper.

Rice husk is a versatile material that has a wide range of potential applications in the construction industry. The following information is provided regarding the application of rice husk in the building industry:

- Because it has a poor heat conductivity, rice husk can be utilized as a material for insulation. It is possible to compress it into panels and use it as hard insulation, or it can simply be utilized as a loose-fill type of insulation.
- Blocks for construction: Rice husk, when combined with cement, can be used to produce building blocks that are not only lightweight but also have high levels of thermal insulation.
- Rice husk may be processed into roofing tiles that are lightweight, resilient, and have strong thermal insulation capabilities. Rice husk is a material that can be used for roofing.
- Rice husk is a material that can be used to absorb sound and can be installed in the ceilings, floors and walls of a building.

Rice husk has a number of benefits that make it a desirable material for use in construction, including the fact that it is inexpensive, abundant, and environmentally friendly. To find the ideal proportions and applications for rice husk in building, however, extensive testing and evaluation of the qualities of the produced materials will be necessary. This is because rice husk has a number of unique properties.

3.3.4 Water hyacinth

Natural fibre has received wide attention from researchers all over the world because of their advantages, which are environmentally friendly and cost-effectiveness. There is different plant fibre among various plant fibre, water hyacinth one source of natural fibre and the plant resource used for animal feed, organic fertilizer, for making paper, bag, basket, and tablecloth. The water hyacinth (*Eichhornia crassipes*) is a free-floating aquatic plant, a native of the Amazon basin and belonging to Pontederiaceae family, with various growth habits under different environmental conditions. Water hyacinth is a noxious weed that has attracted worldwide attention because of its fast spread and congested growth, which lead to serious problems oxygen depletion and reduced water quality large water hyacinth mats prevent the transfer of oxygen from the air to the water surface or decrease oxygen production by other plants and algae. The plant doubles its surface area indoors within two weeks, sometimes in a week. Natural fibre is simple fibre found in plants and animals. Their advantages are high strength, durable, low thermal conductivity, strength related to weight, lightweight, corrosion resistance, and dimensional stability. Water hyacinths are a well-known plant found in aquariums. Because of air-filled sacs, the leaves and stems float in water. Water hyacinth contains low lignin content, thus low lignin content means the hemicellulose and cellulose are easily changed to fermentable sugar, used in the removal of heavy metals such as Zn, Cd, As, Ni, Cr and Hg.



Fig. 3.5 Water hyacinth (Sukarni S et.al, 2019)^[9]

3.4 TESTING OF RAW MATERIALS

The main geotechnical tests are mentioned below:

3.4.1 Determination of water content

As per IS 2720 (Part 2)-1975 The natural water content also called the natural moisture content is the ratio of the weight of water to the weight of the solids in a given mass of soil. This ratio is usually expressed as percentage.

The apparatus of the experiment is non-corrodible air-tight container, Electric oven maintained at the temperature between 105 °C to 110 °C, Desiccator, Balance of sufficient sensitivity.

Clean the container with lid, dry it and weigh it (W_1). Take a specimen of the soil sample in the container from site and weigh with lid (W_2). Keep the container in the oven with lid removed. Dry the specimen to constant weight maintaining the temperature between 105 °C to 110 °C for a period varying with the type of soil but usually 16 to 24 hours. Record the final constant weight (W_3) of the container with dried soil sample.

The water content of the soil sample is calculated from following equation

$$w = \frac{\text{mass of water}}{\text{mass of solids}}$$
$$= \frac{W_2 - W_3}{W_3 - W_1} \times 100$$



Fig. 3.6 Water content determination apparatus
(<https://theconstructor.org>)

3.4.2 Determination of specific gravity

As per IS 2720 (part 4) The knowledge of specific gravity is needed in calculation of soil properties like void ratio, degree of saturation etc.

Pycnometer is a glass jar of 1 litre capacity and fitted with a conical brass cap by means of a screw type cover. The cap has a small hole of 6 mm diameter at its apex. A rubber or fibre washer is placed between the cap and jar to prevent leakage. A sample of wet soil, about 300 gm is taken in pycnometer and weighed. Water is then added to the soil in the pycnometer to make it half full.



Fig. 3.7 Pycnometer used for specific gravity

(www.seminarhunt.com)

The content is thoroughly mixed using a glass rod to remove the entrapped air. More and more water is added and stirring process is continued till the pycnometer is filled flush with the hole in conical cap. The pycnometer is completely emptied and is washed thoroughly and filled with water, flush with the hole in the conical cap.

Note: Take about 300gm of oven dried powdered water hyacinth retained on 0.075 mm IS sieve

3.4.3 Grain size distribution of soil by sieve analysis

As per IS 383:1970 soils having particle larger than 0.075 mm size are termed as coarse-grained soils. In these soils more than 50% of the total material by the mass is larger than 75 μ .



Fig 3.8 Grain size distribution of soil by sieve analysis apparatus

(www.seminarhunt.com)

The apparatus of the experiment are Balances accurate to 1gm and 0.1 gm, Set of IS sieves: 4.75mm, 2.36mm, 1.18mm, 600 μ , 425 μ , 300 μ , 150 μ , 75 micron and pan, Mechanical sieve shake, Sieve brushes and a wire brush. Take about 2000gm of representative dry soil sample. I.S sieves are selected and arranged in the decreasing order as shown in the table. The soil sample is separated into various fractions by sieving through above sieves placed in the above-mentioned order. The weight of soil retained on each sieve is recorded.

Coefficient of curvature is obtained by $C_C = \frac{D_{30}^2}{D_{10} \times D_{60}}$

Uniformity coefficient is obtained by $C_U = \frac{D_{60}}{D_{10}}$

D_{10} is particle size at 10% finer

D_{30} is particle size at 30% finer

D_{60} is particle size at 60% finer

3.4.4 Determination of shrinkage limit of soil

In according with IS 2720 (Part 1)-1972. The shrinkage limit (SL) is the water content where further loss of moisture will not result in any more volume reduction. The shrinkage limit is much less commonly used than the liquid limit and the plastic limit. Shrinkage index (SI or I_s) is defined as the difference between the plastic and shrinkage limits of a soil.

The apparatus of the experiment is Evaporating Dish, Porcelain, about 12 cm in diameter with flat bottom, Spatula, Shrinkage Dish, Straight Edge made of steel, 15 cm in length, Glass cup, Glass plates, Sieves - 425- micron IS sieves.

(a) Preparation of soil sample

Take about 100 gm from a thoroughly mixed portion of the material passing through 425-micron I.S. sieve. Place about 30 gm of the above soil sample in the evaporating dish and thoroughly mix it with distilled water and make a creamy paste. Use water content somewhere around liquid limit.

(b) Filling the shrinkage dish

Coat the inside of the shrinkage dish with a thin layer of Vaseline to prevent the soil sticking to the dish. Fill the dish in three layers by placing approximately $1/3^{\text{rd}}$ of the amount of wet soil with the help of spatula. Tap the dish gently on a firm base until the soil flows over the edges and no apparent air bubbles exist. Repeat this process for 2^{nd} and 3^{rd} layers also till the dish is filled with the wet soil. Strike off the excess soil and make the top of the dish smooth. Wipe off all the soil adhering to the outside of the dish. Weigh immediately, the dish with wet soil and record the weight. Then oven-dry the soil to constant weight at 105°C to 110°C say about 12 to 16 hrs. Remove the dried disk of the soil from oven. Cool it in a desiccator. Then obtain the weight of the dish with dry sample. Determine the weight of the empty dish and record.

(c) Volume of wet soil pat

Determine the volume of shrinkage dish which is evidently equal to volume of wet soil as follows. Place the shrinkage dish in an evaporating dish and fill the dish with mercury till it overflows slightly. Press it with plain glass plate firmly on its top to remove excess mercury. Pour the mercury from the shrinkage dish into a measuring jar and find the volume of the shrinkage dish directly. Record this volume as the volume of the wet soil pat.

(d) Volume of dry soil pat

Determine the volume of dry soil pat by removing the pat from the shrinkage dish and immersing it in the glass cup full of mercury in the following manner. Place the glass cup in a larger one and fill the glass cup to overflowing with mercury. Remove the excess mercury by covering the cup with glass plate with prongs and pressing it. See that no air bubbles are entrapped. Wipe out the outside of the glass cup to remove the adhering mercury. Then, place it in another larger dish, which is, clean and empty carefully. Place the dry soil pat on the mercury. It floats submerge it with the pronged glass plate which is again made flush with top of the cup. The mercury spills over into

the larger plate. Pour the mercury that is displaced by the soil pat into the measuring jar and find the volume of the soil pat directly.



Fig. 3.9 Shrinkage limit determination apparatus
(www.civilengineeringlab.com)

3.4.5 Determination of liquid limit

As per IS 2720 (Part 5) Liquid limit is the water content corresponding to the arbitrary limit between liquid and plastic state of consistency of soil. It is determined as the minimum water content at which the soil is still in the liquid state but has small shearing strength against flowing which can be measured by standard available means. In other words, the liquid limit is the minimum moisture content at which the soil tends to flow as a liquid.

About 120 gm of air-dried soil from thoroughly mixed portion of material passing 425 micron I.S sieve is to be obtained. Distilled water is mixed to the soil thus obtained in a mixing disc to form uniform paste. The paste shall have a consistency that would require 50 to 60 drops of cup to cause closure of standard groove for sufficient length. A portion of the paste is placed in the cup of Casagrande apparatus and spread into portion with few strokes of spatula. Trim it to a depth of 1cm at the point of maximum thickness and return excess of soil to the dish.

The soil in the cup shall be divided by the firm strokes of the grooving tool along the diameter through the centre line of the follower so that clean sharp groove of proper dimension is formed. Lift and drop the cup by turning crank at the rate of two revolutions per second until the two halves of soil cake come in contact with each other for a length of about 1 cm by flow only. The number of blows required to cause the groove close for about 1 cm shall be recorded. A representative portion of soil is taken from the cup for water content determination. Repeat the test with different moisture contents at least three more times for blows between 10 and 40.



Fig. 3.10 Casagrande's apparatus for liquid limit

(www.civilengineeringlab.com)

3.4.6 Determination of plastic limit

The plastic limit (PL) is the water content where soil starts to exhibit plastic behaviour. A thread of soil is at its plastic limit when it is rolled to a diameter of 3 mm or begins to crumble. The apparatus of the experiment is Spatula, Balance, Glass plate, Hot air oven maintained at $105 \pm 10^{\circ}\text{C}$, Moisture Containers.

Take about 20gm of thoroughly mixed portion of the material passing through 425-micron I.S. sieve. Mix it thoroughly with distilled water in the evaporating dish till the soil mass becomes plastic enough to be easily moulded with fingers. Take about 10gms of this plastic soil mass and roll it between fingers and glass plate with just sufficient pressure to roll the mass into a threaded of uniform diameter throughout its length. The rate of rolling shall be between 60 and 90 strokes per minute. Continue rolling till you get a threaded of 3 mm diameter. Knead the soil together to a uniform mass and re-roll. Continue the process until the thread crumbles when the diameter is 3 mm. Collect the pieces of the crumbled thread in airtight container for moisture content determination. Repeat the test to at least 3 times and take the average of the results calculated to the nearest whole number.

3.4.7 Compaction test

As per IS 2720 (Part 8) This method covers the determination of the relationship between the moisture content and density of soils compacted in a mould of a given size with a 2.5 kg rammer dropped from a height of 30 cm.

The apparatus of the experiment is Proctor mould having a capacity of 944 cc with an internal diameter of 10.2 cm and a height of 11.6 cm. The mould shall have a detachable collar assembly and a detachable base plate. A mechanical operated metal rammer having a 5.08 cm diameter face and a weight of 2.5 kg, the rammer shall be

equipped with a suitable arrangement to control the height of drop to a free fall of 30 cm. Sample extruder, a balance of 15 kg capacity. Sensitive balance, Straight edge, Graduated cylinder, Mixing tools such as mixing pan, spoon, towel, spatula etc. Moisture tins.

Take a representative oven-dried sample, approximately 5 kg in the given pan. Thoroughly mix the sample with sufficient water to dampen it to approximately four to six percentage points below optimum moisture content. Weight the proctor mould without base plate and collar. Fix the collar and base plate. Place the soil in the Proctor mould and compact it in 3 layers giving 25 blows per layer with the 2.5 kg rammer falling through. Remove the collar, trim the compacted soil even with the top of the mould by means of the straight edge and weigh. Divide the weight of the compacted specimen by 944 cc and record the result as the wet weight γ_{wet} in grams per cubic centimetre of the compacted soil.

Remove the sample from the mould and slice vertically through and obtain a small sample for moisture determination. Add water in sufficient amounts to increase the moisture content of the soil sample by one or two percentage points and repeat the above procedure for each increment of water added. Continue this series of determination until there is either a decrease or no change in the wet unit weight of the compacted soil.

$$\text{Bulk mass density, } \rho = \frac{M}{V} \text{ gm/ml}$$

where M = mass of compacted soil (gm),

V = volume of the mould (mm).

$$\text{Dry density, } \rho_d = \frac{\rho}{1+w}$$

where, w is the water content



Fig. 3.11 Compaction test apparatus

(www.civilengineeringlab.com)

3.4.8 Hydrometer analysis

As per IS 2720 (Part 4) 1985 A hydrometer usually consists of a sealed hollow glass tube with a wider bottom portion for buoyancy, ballast such as lead or mercury for stability, and a narrow stem with graduations for measuring. The liquid to test was poured into a tall container, often graduated cylinder, and the hydrometer was gently lowered into the liquid until it floats freely. The point at which the surface of the liquid touches the stem of the hydrometer correlates to relative density.

Hydrometer analysis is a widely used method of obtaining an estimate of the distribution of soil particle sizes from 0.075 mm sieve to 0.002 mm. The data are presented on a semi log plot of percent finer vs particle diameter and may be combined with the data from a sieve analysis of the material retained on the 75 μ . This test is done when more than 20 % pass through 75 - μ and 90 % or more passes 4.75 mm sieve. The hydrometer makes use of Archimedes principle: a solid suspended in a fluid is buoyed by a force equal to the weight of the fluid displaced by the submerged part of the suspended solid. The lower the density of the fluid, the deeper a hydrometer of a given weight sinks, the stem is calibrated to give a numerical reading. The hydrometer analysis is based on Stokes Law, which gives the relationship among the velocity of fall of spheres in a fluid, the diameter of the sphere, the specific weights of the sphere and of the fluid, and the fluid viscosity.

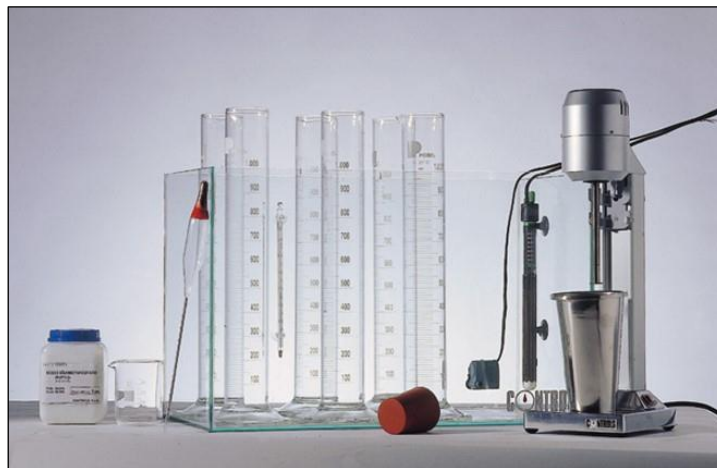


Fig. 3.12 Hydrometer analysis apparatus

(www.blogedu.com)

3.5 TESTING OF SPECIMEN

The main tests which are conducted on the specimen are given below.

3.5.1 Compressive strength

As per IS 3495 (Part 1), IS 1077(1970) The compressive strength of the brick is determined to analyse the strength characteristics.

Apparatus: A compression testing machine, the compression plate of which shall have a ball seating in the form of portion of a sphere the centre of which coincides with the centre of the plate, shall be used.

Preconditioning: Remove unevenness observed in the bed faces to provide to smooth and parallel faces by grinding. Immerse in water at room temperature for 21 hours. Remove the specimen and drain out any surplus moisture at room temperature. Store under the damp jute bags for 24 hours followed by immersion in clean water for 3 days. Remove, and wipe out any traces of moisture.

Procedure: Place the specimen with flat faces horizontal facing upwards and carefully centred between plates of the testing machine. Apply load axially at a uniform rate of 14 N/mm^2 (140 kgf/cm^2) per minute till failure occurs and note the maximum load at failure. The load at failure shall be the maximum load at which the specimen fails to produce any further increase in the indicator reading on the testing machine. Compressive strength in N/mm^2 =

$$\frac{\text{maximum load at failure in N(kgf)}}{\text{average area of the bed face in mm}^2}$$

Table 3.2 Classes of common bricks

Class designation	Average compressive strength not less than N/mm^2
35	35
30	30
25	25
20	20
17.5	17.5
10	10
7.5	7.5
5	5
3.5	3.5

Note: Three specimens of each specified proportions of clay, red mud, rice husk and water hyacinth are tested by placing them at compression testing machine and by applying pressure axially until it breaks. Average results are considered as the compressive strength.

3.5.2 Water absorption test

As per IS 3495 (Part 2) Water absorption test on bricks is conducted to determine durability properties of bricks. Pores affect the water absorption property of bricks. More the pores, more will be the water absorption. Bricks, which have water absorption less than 3% can be called as vitrified. Apart from compressive strength, water absorption is an important characteristic.

Apparatus: A sensitive balance capable of weighing within 0.1 percent of the mass of the specimen; and a ventilated oven.

Preconditioning: Dry the specimen in a ventilated oven at a temperature of 105 °C to 115 °C till it attains significantly constant mass. Cool the specimen to room temperature and obtain its weight (W_1). Specimen warm to touch shall not be used for the purpose.

Procedure: In this test, first, bricks are weighed in dry condition (W_1) followed by full submersion in water for 24 hours. After immersion in water for 24 hours, the bricks are weighed again in wet condition (W_2). The difference of weight is considered as the water absorbed by the bricks. Then water absorption is determined in percentage. Greater the quality, lesser will be the water absorption. The brick is considered as good quality brick if it does not consume more than 20% water of its own weight.

$$\text{Water absorption} = \frac{W_2 - W_1}{W_1} \times 100$$

3.5.3 Efflorescence test

As per IS 3495 (Part 3) Efflorescence is the process of analysing the presence of alkaline substances in bricks.

Apparatus: A shallow flat bottom dish containing sufficient distilled water to completely saturate the specimen. The dish shall be made of glass, porcelain or glazed stone ware and of size 180mm x 180mm x 40 mm depth.

Procedure: : Place the end of the bricks in the dish, the depth of immersion in water being 25 mm. Place the whole arrangement in a warm ventilated room of temperature of about 20 to 30°C until all the water in the dish is absorbed by the specimens. And the excess water evaporates. Cover the dish containing the brick with suitable glass

cylinder so 27 that excessive evaporation from the dish may not occur. When the water has been absorbed and bricks appear to be dry, place a similar quantity of water in the dish and allow it to evaporate. Examine the bricks for efflorescence after the second evaporation. Efflorescence shall be reported as 'nil', 'slight', 'moderate', 'heavy' or 'serious'.

- (a). Nil- When there is no perceptible deposit of efflorescence.
- (b). Slight- When not more than 10 percent of the exposed area of the brick is covered with a thin deposit of salts.
- (c). Moderate- When there is a heavier deposit than under slight and covering up to 50 percent of the exposed area of the brick surface but unaccompanied by powdering or flaking of the surface.
- (d). Heavy- When there is a heavy deposit of salts covering 50 percent or area of the brick surface but unaccompanied by powdering or flaking of the surface.
- (e). Serious- When there is a heavy deposit of salts accompanied by powdering or flaking of the exposed surfaces.

Note: If the efflorescence is greater than 50% such condition is treated as serious, and brick is rejected.

3.5.4 Hardness test

Hardness test is the process of evaluating the resistance against abrasion. In this test a scratch is tried to make on the brick surface with the help of a fingernail. If no impression is left on the surface, the brick is enough hard.

3.5.5 Soundness test

Soundness test is done to analyse the resistance of brick against impact load. In this test two bricks are taken and they have stuck each other, the brick should not break and produces a clear ringing sound which implies they are of good quality.

3.5.6 Impact test

In this test, take a brick and drop it from a one-metre height. A good quality brick should not break at all. If the bricks are broken that means its impact value is low and rejected.

CHAPTER 4

STUDY ON CLAY BLOCKS

4.1 GENERAL

In this chapter we are discussing about material testing, casting, curing and testing of clay blocks.

4.2 MATERIAL USED

The material used for analysis is clay. Clay was collected from a locally operated brick factory in Puthiyara. They were powdered and sieved. The material properties are discussed in Table 4.1 onwards.



Fig 4.1 Powdered clay

Table 4.1 Test results of clay

Sample	Test conducted	Values	Permissible limit
Clay	Water content	14.165	10-35
	Specific gravity	2.62	2.60-2.80
	Shrinkage limit	30	20-40
	Plastic limit	25	15-100

Table 4.2 Liquid limit test result of clay

Moisture Content	No. of blows
29	52
35	36
42	25
52	14
74	3

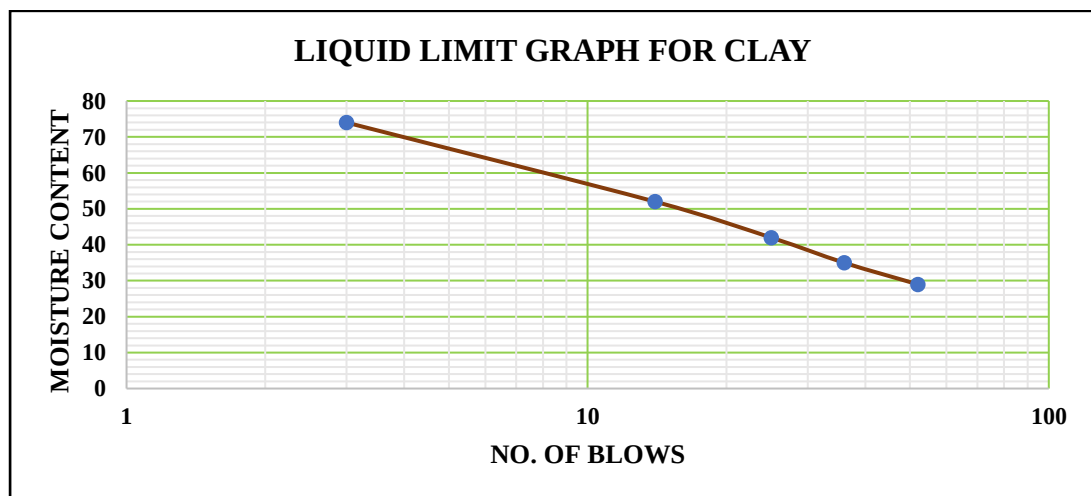


Fig. 4.2 Liquid limit graph for clay

From the graph the liquid limit value obtained is 42% and plasticity index is 17% which comes under non-plasticity slit, which is a type of alluvial clay. The flow index and toughness index are 59.57 and 0.28 respectively.

Table 4.3 Compaction test values of clay

Moisture content (%)	Bulk density (gm/cc)	Dry density (gm/cc)	Optimum moisture content (%)
8	1.32	1.38	24
16	1.59	1.41	
24	1.64	1.44	
32	1.61	1.39	

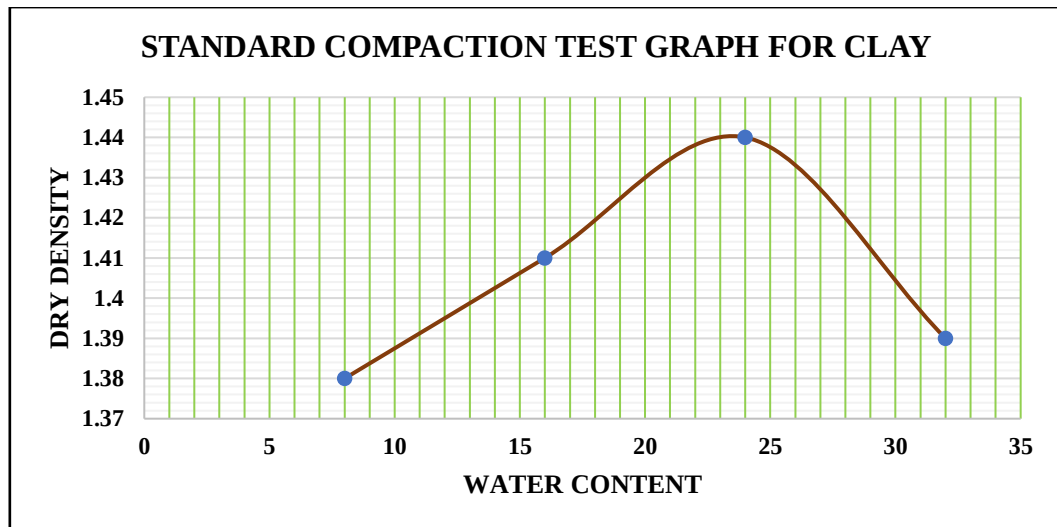


Fig. 4.3 Compaction curve for clay

The maximum dry density and optimum moisture content for clay are 1440 kg/m³ and 24%.

Table 4.4 Calibration curve for clay

Hydrometer reading (Rh)	Effective depth $t \geq 2$ min	Effective depth $t \leq 2$ min
0	16.74	17.5
10	15.74	16.5
20	14.74	15.5
30	13.74	14.5
40	12.74	13.5
50	11.74	12.5
60	10.74	11.5
70	9.74	10.5
80	8.74	9.5
90	7.74	8.5
100	6.74	7.5

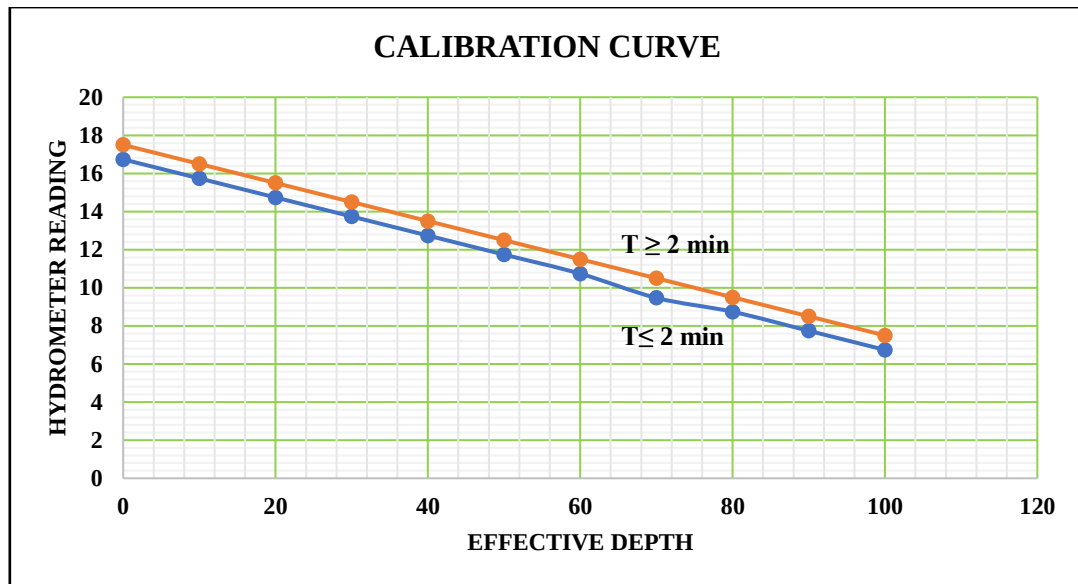


Fig. 4.4 Calibration curve of clay

Table 4.5 Grain size distribution values of clay

Time elapsed (min)	% finer N
0.5	41.75
1	40.15
2	38.54
4	37.9
8	35.33
15	33.72
30	32.12
60	31.47

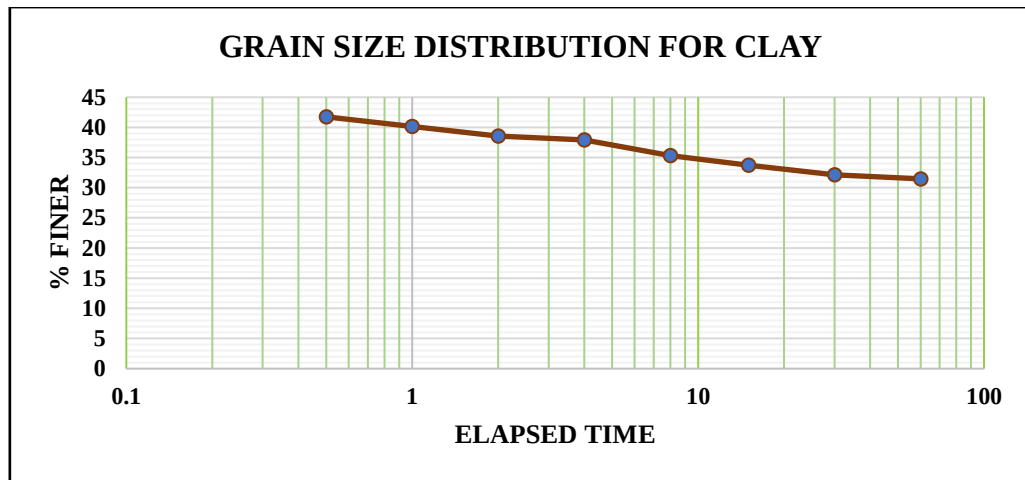


Fig. 4.5 Grain size distribution for clay

4.3 CASTING OF SPECIMEN

A variety of brick combinations are prepared after the material property analysis. The brick specimen was downscaled from the commercial size of 230mm x 110mm x 79mm (BIS 1077, 1992) to 120mm x 120mm x 60mm, due to the unavailability of other raw materials desired. As a result, they were pressed under 40-50 MPa, the samples are oven dried at 105°C for 24 hours, followed by a 2-hour blast furnace firing blocks.

4.4 TESTING OF SPECIMEN

Water absorption, efflorescence, soundness, hardness, impact resistance and compressive strength analysis were done on the specimen according to Indian testing standards. To ensure the accuracy, three trials were performed on each test and the mean value was selected.

Table 4.6 Test results of clay blocks

S.No	1
Sample	Clay (100%)
Size	120x120x60
Compressive strength (N/mm ²)	6.64
Water absorption	15%
Efflorescence	nil
Hardness	no
Soundness	high
Impact	high



Fig. 4.6 Weighting of clay block



Fig. 4.7 Compression testing of clay block

4.5 RESULT AND DISCUSSION

The compressive strength of clay block obtained is 6.64 N/mm^2 , water absorption is 15% which is under the permissible limit of 20% because of good cohesion and adhesion. There is no efflorescence present in the specimen and having high impact strength. The compressive strength value of 6.64 N/mm^2 is taken for final comparison.

CHAPTER 5

STUDY ON CLAY BLOCK IN PARTIAL REPLACEMENT WITH RICE HUSK

5.1 GENERAL

In this chapter we are discussing about material testing, casting, curing and testing of clay blocks partially replaced with rice husk.

5.2 MATERIALS USED

The materials used were clay and rice husk. The materials properties of clay have been discussed in section 4.2. Table 5.1 onwards shows the various test and results of RH. RH was collected from Omalloor as well as Puthiyara. Powdering was done as per the conventional purpose.



Fig. 5.1 & 5.2 Collection of rice husk

Table 5.1 Test results of rice husk

Sample	Test conducted	Values	Permissible limit
Rice husk	Water content	13.59	10-15
	Specific gravity	2.24	2.11-2.27
	Plastic limit	31.25	31.20-31.67

Table 5.2 Liquid limit test result of rice husk

Moisture Content	No. of blows
78	48
82	33
90	25
100	19
120	10

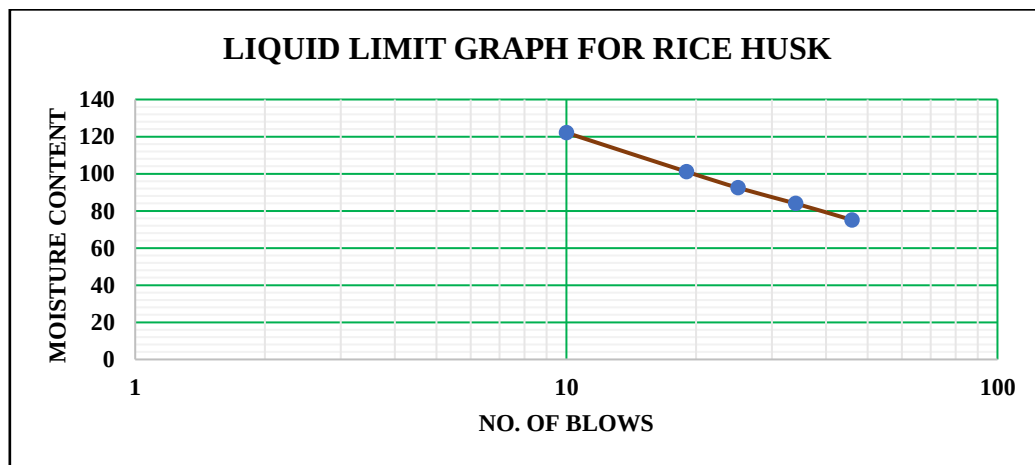


Fig. 5.3 Liquid limit graph for rice husk

From the graph the liquid limit value obtained is 91.25% and plasticity index is 59.73%. The flow index and toughness index be 83.83 and 0.71 respectively.

5.3 CASTING OF SPECIMEN

Size of mould and reason for downscaling was mentioned in section 4.3. Clay brick infused with rice husk at rates of 2%, 5%, 8%, 11% and 14% by weight.

5.4 TESTING OF SPECIMEN

Water absorption, efflorescence, soundness, hardness, impact resistance and compressive strength analysis were done on the specimen according to Indian testing standards. To ensure the accuracy, three trials were performed on each test and the mean value was selected.

Table 5.3 Test results of clay and rice husk

S.No	1	2	3	4	5
Sample	RH (2%)	RH (5%)	RH (8%)	RH (11%)	RH (14%)
Size	120x120x60	120x120x60	120x120x60	120x120x60	120x120x60
Compressive strength (N/mm ²)	3.31	3.49	3.67	4.51	4.21
Water absorption	20.51	20.12	19.61	18.1	18.6
Efflorescence	nil	nil	nil	low	low
Hardness	no	no	no	yes	yes
Soundness	moderate	moderate	high	high	high
Impact	low	low	moderate	high	moderate

5.5 RESULT AND DISCUSSION

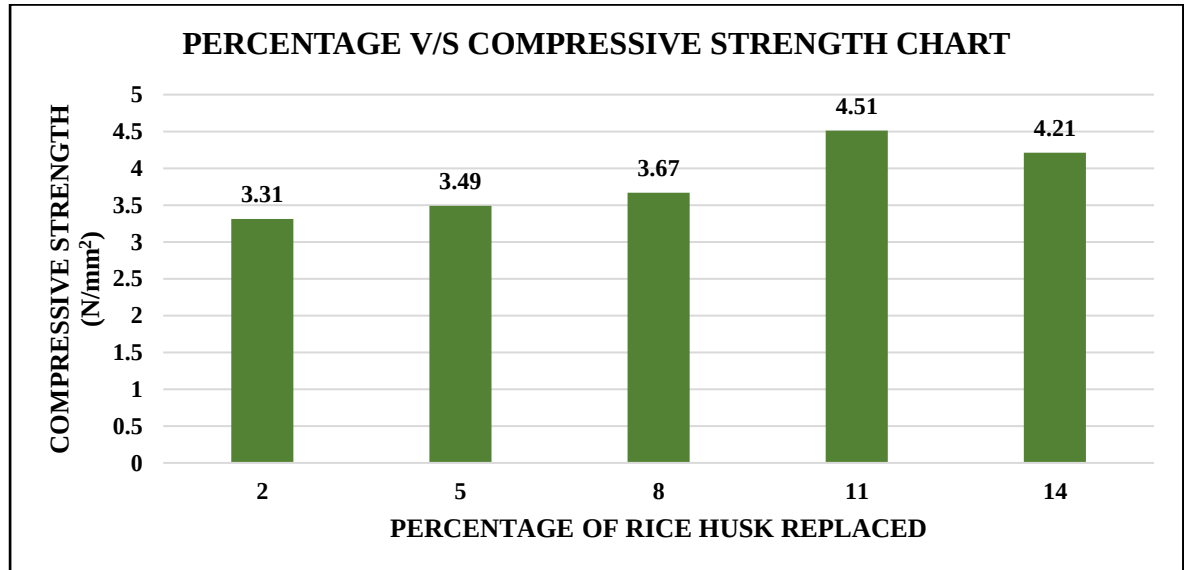


Fig. 5.4 Compressive strength comparison of clay partially replaced with rice husk

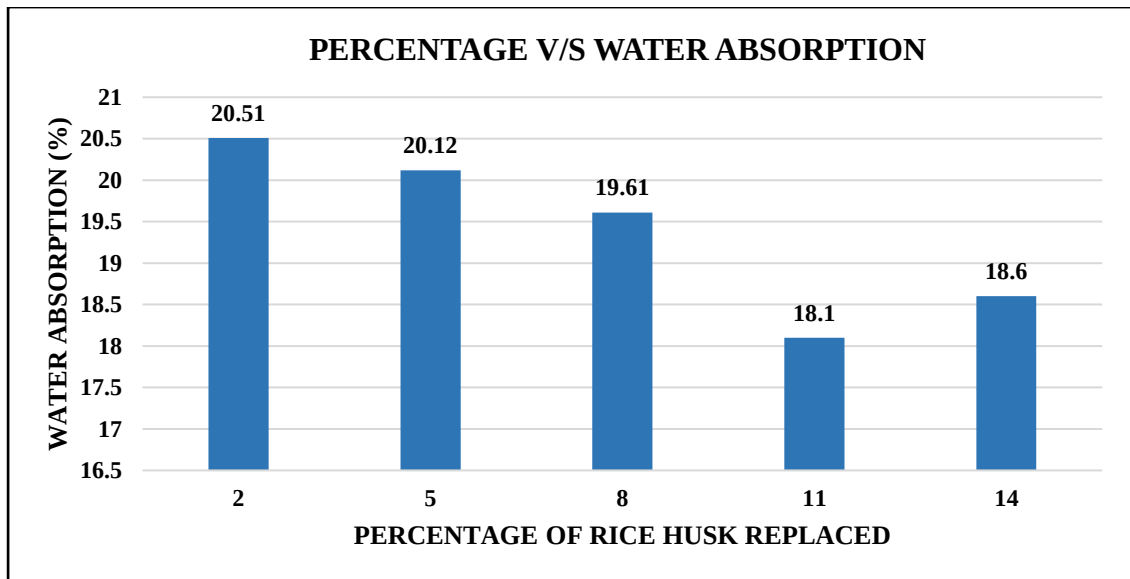


Fig. 5.5 Water absorption comparison of clay partially replaced with rice husk

The compressive strength increases with increase in percentage replaced up to 11% (4.51 N/mm^2) and then decreases. The silica content of rice husk help in increasing the compressive strength. The water absorption is initially found at the range slighter that the permissible limit of 20% and then decreases, there is no efflorescence up to 8% and there is a slight presence at 11% and 14%. Hardness and soundness are also present and there is high impact resistance in 14%. Therefore, the optimum percentage obtained is 11% replacement and it can be taken for final consideration.

CHAPTER 6

STUDY ON CLAY BLOCK IN PARTIAL REPLACEMENT WITH RED MUD

6.1 GENERAL

In this chapter we are discussing about material testing, casting, curing and testing of clay blocks partially replaced with red mud.

6.2 MATERIALS USED

The materials used were clay and red mud. The materials properties of clay have been discussed in section 4.2. Table 6.1 onwards shows the various test and results of RM. RM was collected from a local supplier. They were weighed and sieved through 75-micron sieve.



Fig. 6.1 Red mud

Table 6.1 Test results of red mud

Sample	Test conducted	Values	Permissible limit
Red mud	Water content	44.57	25-60
	Specific gravity	3.075	2.85-3.45
	Shrinkage limit	12.75	10-15
	Plastic limit	32	30-35

Table 6.2 Grain size distribution of red mud by sieve analysis test values.

Sieve size	Mass of soil retained	% of soil retained	Cumulative % of soil retained	% Finer
4.75 mm	2	0	0	100
2.36 mm	326	8.33	8.33	91.67
1.18 mm	472	8.89	17.22	82.78
600 μ	252	16.67	33.89	66.11
425 μ	356	16.67	50.56	49.44
300 μ	94	16.33	66.89	33.1
150 μ	428	21.99	88.88	11.11
75 μ	41	10.15	99.03	0.97
pan	29	0.97	100	0.00

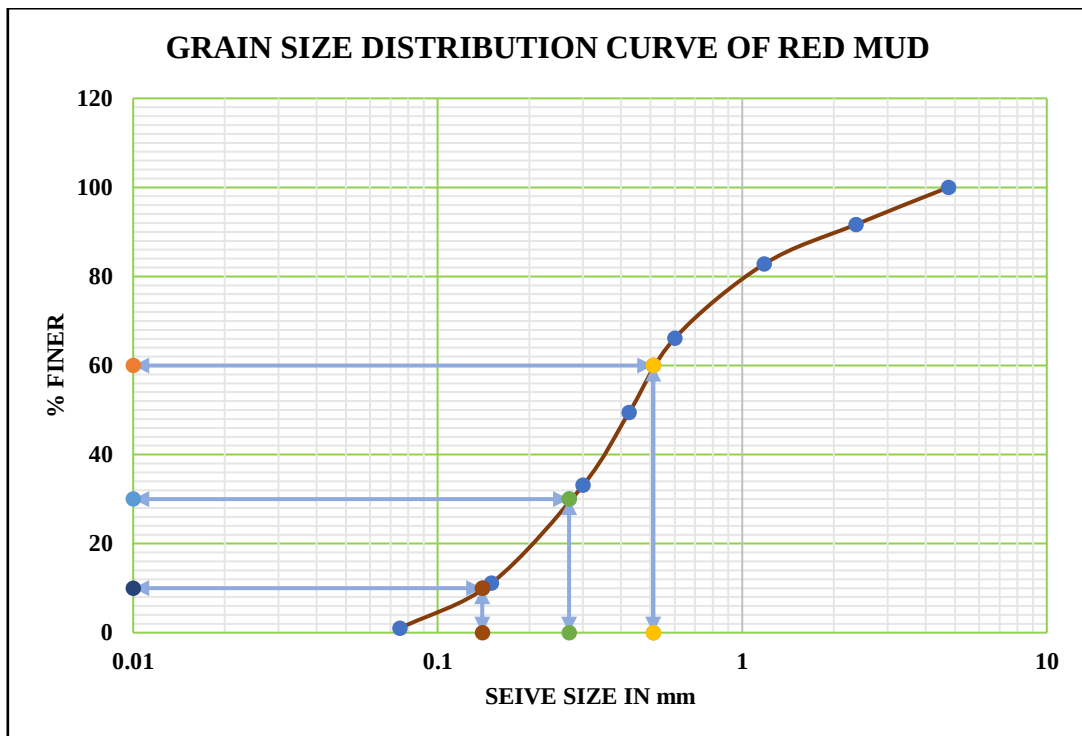


Fig. 6.2 Grain size distribution curve of red mud

Table 6.3 Result of grain size distribution curve

D ₁₀	0.14
D ₃₀	0.27
D ₆₀	0.51
C _C	3.64
C _U	1.02

Particle size distribution mud indicated grains are fine and it is well graded.

Table 6.4 Liquid limit test result of red mud

Moisture Content	No. of blows
28	52
35	34
48	14
58	7
74	2

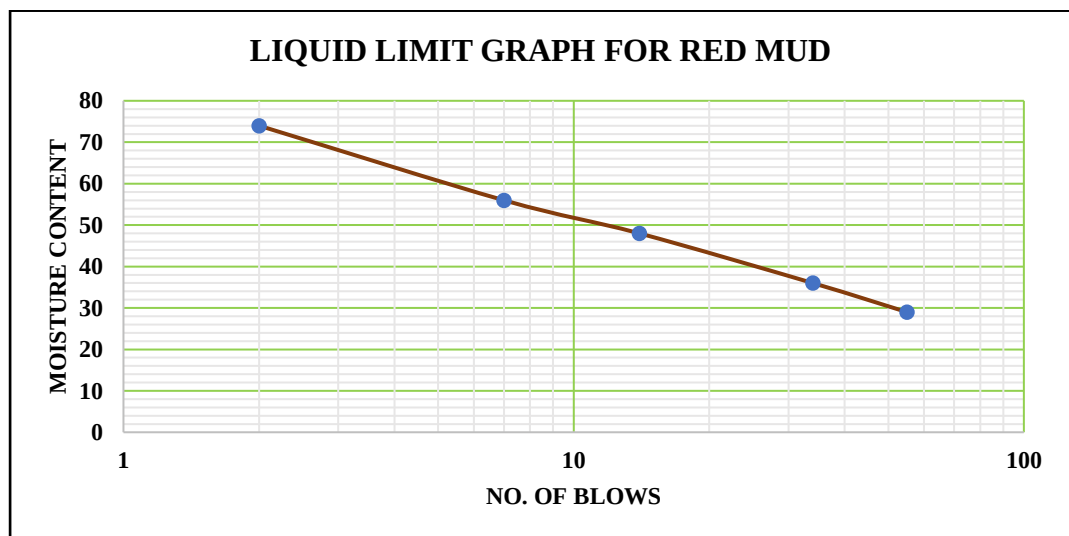


Fig. 6.3 Liquid limit graph for red mud

From the Atterberg's limits it is concluded that the plasticity index of the red mud is 13, liquid limit is found to be 45%. Flow index and toughness index are 25.12 & 0.51 respectively.

Table 6.5 Compaction test values of red mud

Moisture content (%)	Bulk density (gm/cc)	Dry density (gm/cc)	Optimum moisture content (%)
5	1.01	1.1	35
10	1.13	1.12	
15	1.21	1.15	
20	1.32	1.19	
25	1.41	1.20	
30	1.58	1.25	
35	1.64	1.29	
40	1.55	1.24	

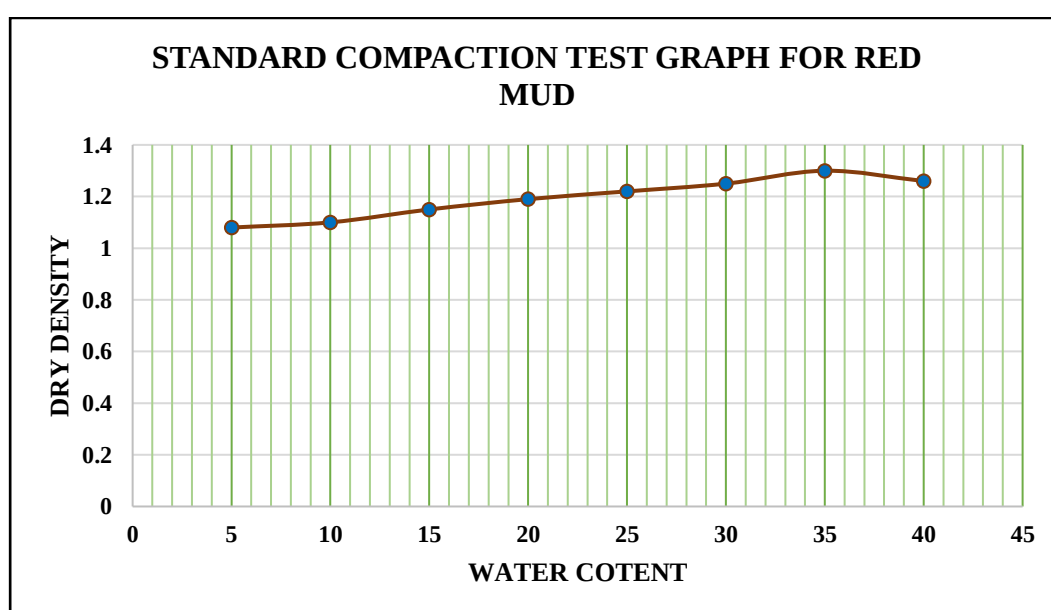


Fig. 6.4 Compaction curve of red mud

The maximum dry density and optimum moisture content of the red mud is 1290 kg/m^3 and 35% respectively.

6.3 CASTING OF SPECIMEN

Size of mould and reason for downscaling was mentioned in section 4.3. Additionally, clay bricks are prepared, pressed under 40-50 MPa and dried under shade for 24 hours using red mud 5%, 15%, 25%, 35% and 45%.



Fig. 6.5 & 6.6 Casting of blocks with clay and red mud.

6.4 TESTING OF SPECIMEN

Water absorption, efflorescence, soundness, hardness, impact resistance and compressive strength analysis were done on the specimen according to Indian testing standards. To ensure the accuracy, three trials were performed on each test and the mean value was selected.

Table 6.6 Test results of clay and red mud

S.No	1	2	3	4	5
Sample	RM (5%)	RM (15%)	RM (25%)	RM (35%)	RH (45%)
Size	120x120x60	120x120x60	120x120x60	120x120x60	120x120x60
Compressive strength (N/mm ²)	5.9	6.8	7.6	6.6	4.1
Water absorption	14	15.6	16.4	17.2	17.9
Efflorescence	nil	nil	nil	low	low
Hardness	no	no	no	no	yes
Soundness	high	high	high	moderate	moderate
Impact	high	high	high	high	moderate

6.5 RESULT AND DISCUSSION

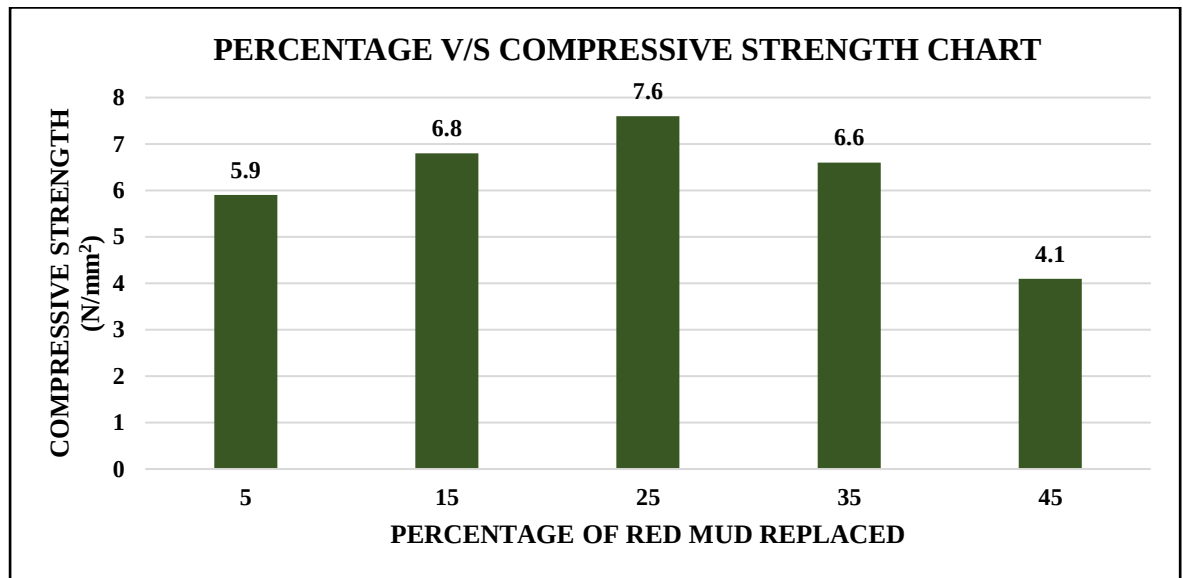


Fig. 6.7 Compressive strength comparison of clay partially replaced with red mud

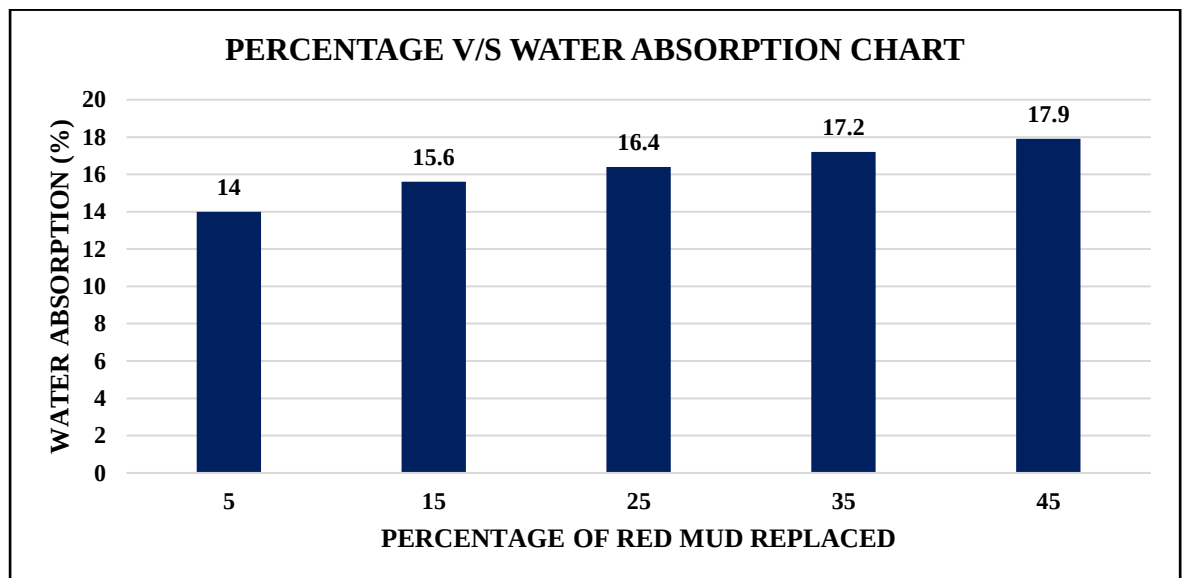


Fig. 6.8 Water absorption comparison of clay partially replaced with red mud

The compressive strength value obtained at 25% replacement with red mud is 7.6 N/mm², which is found to be greater than the value taken for comparison. Pozzolanic property help in improving compressive strength. After analysing the water absorption, it is found to be at the permissible limit of 20% and for 25% replacement it is 16.4%. There is slight presence of efflorescence. Therefore, the optimum percentage obtained is 25% replacement and it can be taken for final consideration.

CHAPTER 7

STUDY ON CLAY BLOCK IN PARTIAL REPLACEMENT WITH WATER HYACINTH

7.1 GENERAL

In this chapter we are discussing about material testing, casting, curing and testing of clay blocks partially replaced with water hyacinth.

7.2 MATERIAL USED

The materials used were clay and red mud. The materials properties of clay have been discussed in section 4.2. Table 7.1 onwards shows the various test and results of WH. WH was collected from Haripad, Alapuzha.



Fig. 7.1 Collection of water
hyacinth



Fig. 7.2 Washed and cleaned water
hyacinth



Fig. 7.3 Water hyacinth is cutting
small pieces



Fig. 7.4 Water hyacinth is kept for
drying



Fig. 7.5 Water hyacinth fully dried

Table 7.1 Test results of water hyacinth

Sample	Test conducted	Values	Permissible limit
Water hyacinth	Water content	12.62	10-15
	Specific gravity	1.081	0.96-1.58
	Plastic limit	52	42-56

Note : Dried and powdered water hyacinth retained on 0.075 mm IS sieve is taken for analysis.

Table 7.2 Liquid limit test result of water hyacinth

Moisture Content	No. of blows
62	38
74	30
84.06	25
102	16
124	12

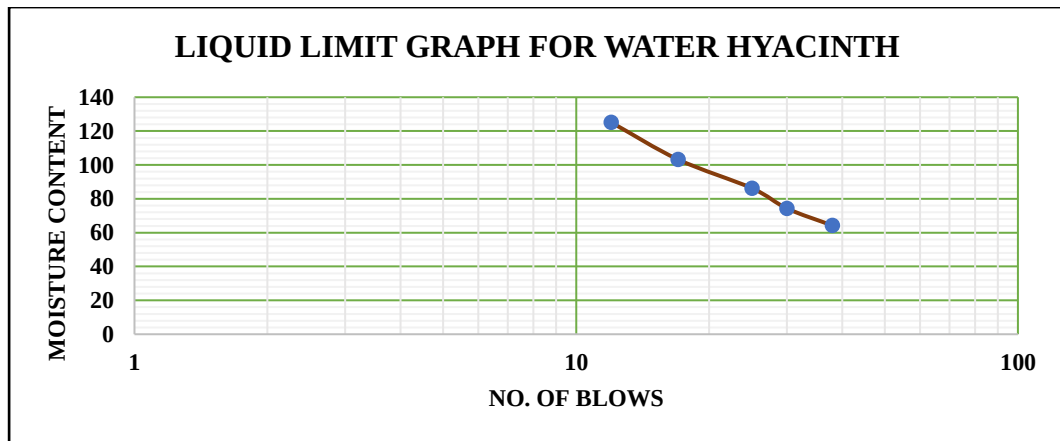


Fig. 7.6 Liquid limit graph for water hyacinth

From the graph the liquid limit value obtained is 84.06% and plasticity index is 32.06. The flow index and toughness index are 56.78 and 0.56 respectively.

Table 7.3 Compaction test values of water hyacinth

Moisture content (%)	Bulk density (gm/cc)	Dry density (gm/cc)	Optimum moisture content (%)
6	0.89	0.58	13
8	0.92	0.591	
10	0.978	0.624	
12	1.01	0.653	
14	0.98	0.612	

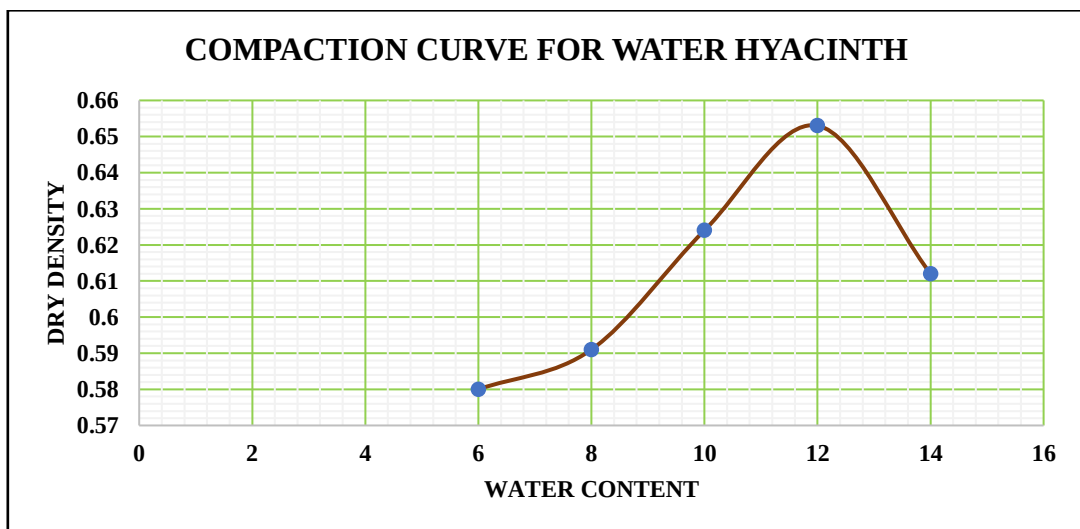


Fig. 7.7 Compaction curve of water hyacinth

The maximum dry density and optimum moisture content of the water hyacinth is 653 kg/m³ and 13% respectively.

7.3 CASTING OF SPECIMEN

Size of mould and reason for downscaling was mentioned in section 4.3. Water hyacinth added at rate of 1%, 3%, 5%, 7% and 10% by weight of clay. Due to unavailability of water hyacinth at conventional purposes, downscaling was done.

7.4 TESTING OF SPECIMEN

Water absorption, efflorescence, soundness, hardness, impact resistance and compressive strength analysis were done on the specimen according to Indian testing standards. To ensure the accuracy, three trials were performed on each test and the mean value was selected.

Table 7.4 Test results of clay and water hyacinth

S.No	1	2	3	4	5
Sample	WH (1%)	WH (3%)	WH (5%)	WH (7%)	WH (10%)
Size	120x120x60	120x120x60	120x120x60	120x120x60	120x120x60
Compressive strength (N/mm ²)	3.9	4.6	5.1	4.7	3.4
Water absorption	18.73	18.8	19.4	20.31	21.6
Efflorescence	nil	nil	moderate	high	high
Hardness	no	no	no	no	yes
Soundness	high	high	moderate	moderate	low
Impact	high	high	high	high	moderate

7.5 RESULT AND DISCUSSION

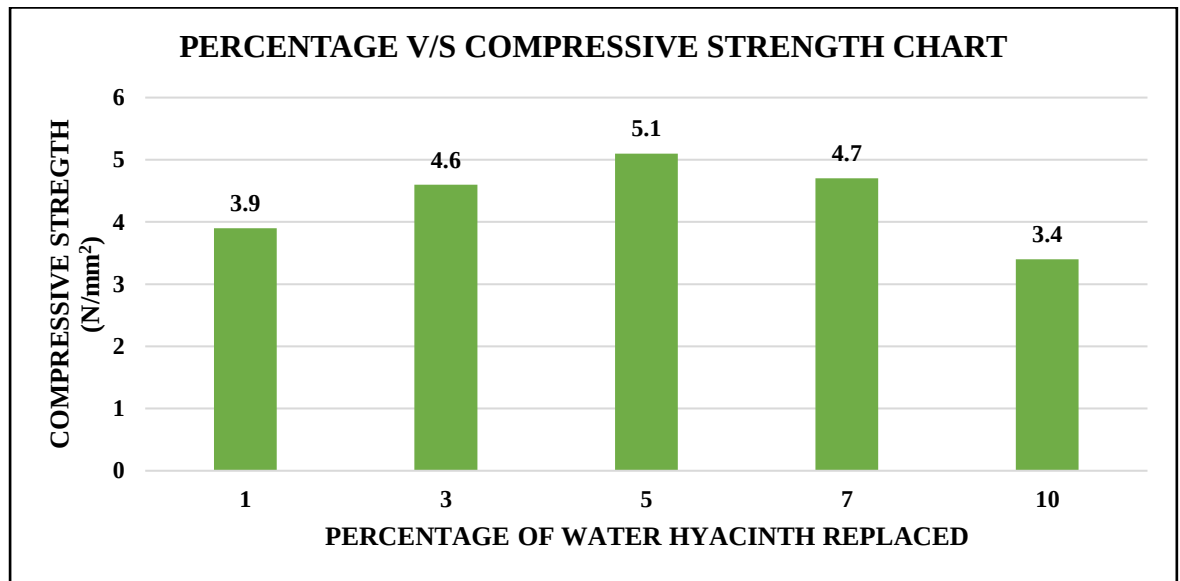


Fig. 7.8 Compressive strength comparison of clay partially replaced with water hyacinth

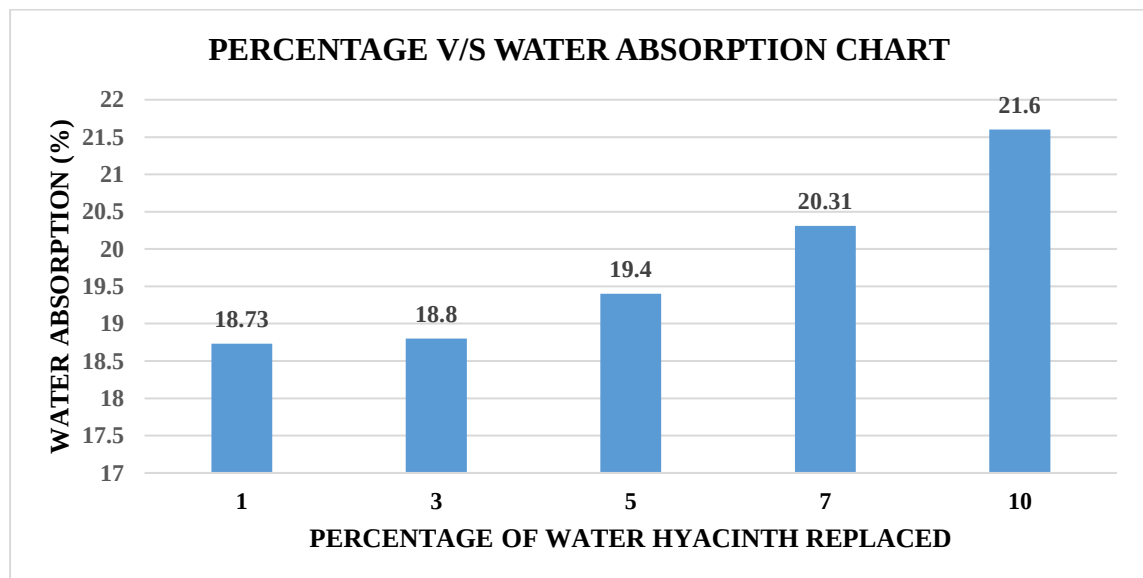


Fig. 7.9 Water absorption comparison of clay partially replaced with water hyacinth

The maximum compressive strength value obtained is 5.1 N/mm² at 5% of water hyacinth replacement, similarly water absorption value obtained is 19.41% for 5% and then increases less than the permissible limit of 20%. There is a moderate value of efflorescence obtained. Therefore, the optimum percentage obtained is 5% replacement and it can be taken for final consideration.

CHAPTER 8

STUDY ON CLAY BLOCK WITH RED MUD, RICE HUSK AND WATER HYACINTH

8.1 GENERAL

In this chapter we are discussing about material testing, casting, curing and testing of blocks at optimum percentage, i.e., best combination selected from each chapter.

8.2 MATERIAL USED

Clay, red mud, rice husk and water hyacinth. Sources of each raw materials were mentioned in previous chapters. The materials properties of clay, rice husk, red mud and water hyacinth have been discussed in section 4.2, 5.2, 6.2 and 7.2 respectively.

8.3 CASTING OF SPECIMEN

Size of mould and reason for downscaling was mentioned in section 4.3. The optimum is obtained as 11% rice husk, 25% red mud and 5% water hyacinth and 59% of clay.



Fig. 8.1 Casting of specimen



Fig.8.2 Casted specimen

8.4 TESTING OF SPECIMEN

Water absorption, efflorescence, soundness, hardness, impact resistance and compressive strength analysis were done on the specimen according to Indian testing standards. To ensure the accuracy, three trials were performed on each test and the mean value was selected.

Table 8.1 Test result of clay, rice husk, red mud and water hyacinth

S.No	1
Sample	Clay (59%), RH (11%), RM (25%) & WH (5%)
Size	120x120x60
Compressive strength (N/mm ²)	6.96
Water absorption	16.75
Efflorescence	nil
Hardness	no
Soundness	high
Impact	high

8.5 RESULT AND DISCUSSION

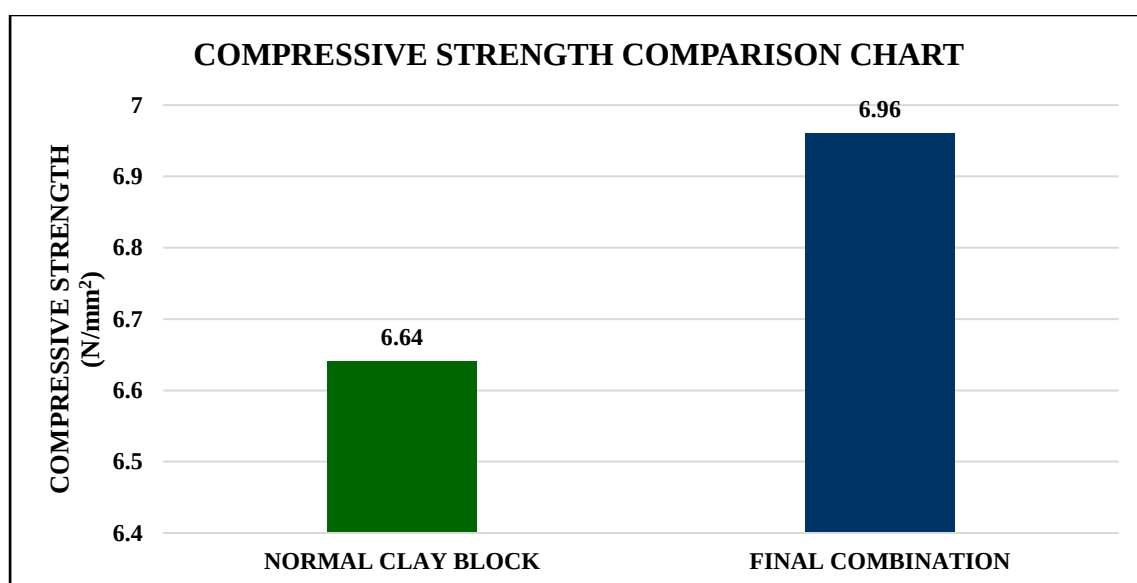


Fig. 8.3 Compressive strength comparison of normal clay block and final combination

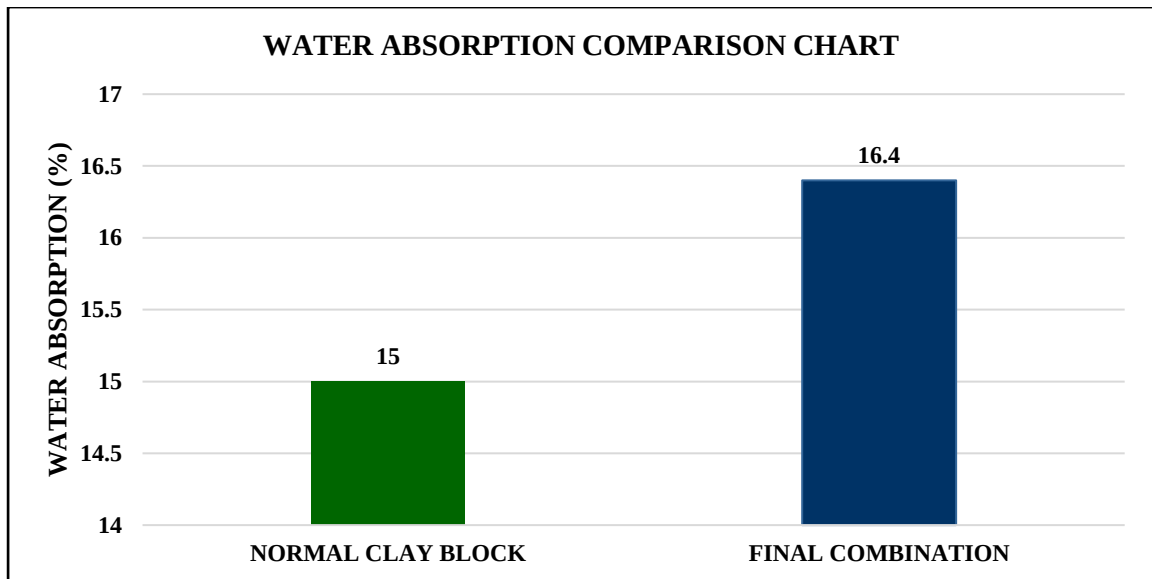


Fig. 8.4 Water absorption comparison of normal clay block and final combination

The optimum is obtained as 11% rice husk, 25% red mud and 5% water hyacinth and 59% clay. The compressive strength value obtained is 6.96 N/mm^2 which is greater than 6.64 N/mm^2 taken for standard comparison. Water absorption is 16.75% which is in the permissible limit of 20%. Efflorescence is not identified, there is no hardness obtained and having high soundness and impact strength.

CHAPTER 9

CONCLUSIONS

Clay blocks are manufactured by using novel materials such as red mud, rice husk, and water hyacinth to reduce environmental problems. These materials have the potential to improve the sustainability of the process of creating blocks as well as to lessen the impact that the process has on the environment.

The raw materials used for the production of blocks are cheaper and locally available. Clay has a property known as plasticity, which enables it to be moulded and formed when it is wet and then to retain its shape when it is dry. Red mud reduces the energy consumption in production but also improves strength and resist sulfate attack. The silica content in RH can be utilized in the construction world as an added material to increase the compressive strength. Water hyacinth has high strength, durable, low thermal conductivity, strength related to weight, lightweight, corrosion resistance, and dimensional stability.

Soil data had been obtained and analysed with the scope of the study. All soil information were obtained from laboratory tests. Data acquired for analysis are from liquid limit, plasticity index, optimum moisture content, sieve analysis, maximum dry density, specific gravity.

Based on the analysis carried out, the conclusions of the study can be summarized as follows:

- Highest value of water content is 44.57% for red mud and lowest value is 12.62% for water hyacinth.
- Pycnometer test is adopted to find out specific gravity. Specific gravity of RM is 3.075. Oven dried and powdered WH retained on 0.075 IS sieve is selected for the same.
- The grain size distribution of red mud is an S-shaped curve and is fine-grained.
- From the Atterberg's limits it is concluded that the plasticity index of the red mud is 13. So, according to the IS classification based on plasticity, the soil falls under ML. Means it is silt with low Permeability. The plasticity index of clay is found to be 32% which comes under non-plasticity slit, which is a type of alluvial clay.
- The maximum dry density increases with the increase in the moisture content up to optimum moisture content and then decreases.

- The maximum dry density and optimum moisture content of the red mud is 1290 kg/m³ and 35% respectively and for clay they are 1240 kg/m³ and 24% respectively.
- As per IS specifications the standard size of block is 230mm x 110mm x 79mm but due to unavailability of water hyacinth the block was rescaled into 120mm x 120mm x 60mm.
- Water was taken up to 13% for clay block and for varying proportion it was taken at a rate to produce a fine paste.
- The compressive strength of 100% clay block is 6.64 N/mm², which is taken for comparison. Water absorption were also to be in the permissible limit of 20%. The blocks have high soundness and impact resistance, and efflorescence is not present in the sample.
- In clay and rice husk combination compressive strength increases with increase in percentage replaced up to 11% and reaches a maximum value of 4.51 N/mm² and then decreases. Therefore, 11% replacement is considered to be optimum, and it can be taken for final consideration. Silica content in rice husk help in increasing the compressive strength.
- The compressive strength value obtained at 25% replacement with red mud is 7.6 N/mm², which is found to be the maximum value and pozzolanic property helps in improving the compressive strength. Efflorescence is found at 35% and 45% replacement.
- The maximum compressive strength value obtained is 5.1 N/mm² at 5% of water hyacinth replacement, similarly water absorption value obtained is 19.41% for 5%.
- From the compressive strength comparison, the optimum percentage obtained is 11% rice husk, 25% red mud, 5% water hyacinth and 59% of clay. The average compressive strength obtained is 6.96 N/mm², which is greater than 6.64 N/mm².
- Water absorption is 16.75% which is in the permissible limit of 20%. Efflorescence is not identified, there is no hardness obtained and having high soundness and impact strength.
- The obtained compressive strength is 6.96 N/mm² (70.97 kg/cm²), hence second-class brick obtained. They can be of low-quality bricks and they can be used for hidden masonry, pitching works and no traffic category.

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APPENDIX-A

A.1 GENERAL

This chapter deals with the mix design.

A.2 MIX DESIGN

A.2.1 Mix design for clay block

Quantity for 9 specimens = 14.95 kg

OMC is 13%.

Therefore, quantity of water = 1.94 litre.

A.2.2 Mix design for clay block partially replaced with rice husk

Quantity for 2% of rice husk replaced with clay, for 9 specimens.

Quantity of rice husk = 0.299 kg

Quantity of clay = 14.651 kg

OMC is 12.85

Therefore, quantity of water = 1.92 litre.

Quantity for 5% of rice husk replaced with clay, for 9 specimens.

Quantity of rice husk = 0.74 kg

Quantity of clay = 14.202 kg

OMC is 12.21

Therefore, quantity of water = 1.82 litre.

Quantity for 8% of rice husk replaced with clay, for 9 specimens.

Quantity of rice husk = 1.196 kg

Quantity of clay = 13.754 kg

OMC is 12.14

Therefore, quantity of water = 1.81 litre.

Quantity for 11% of rice husk replaced with clay, for 9 specimens.

Quantity of rice husk = 1.64 kg

Quantity of clay = 13.30 kg

OMC is 12.09

Therefore, quantity of water = 1.8 litre.

Quantity for 14% of rice husk replaced with clay, for 9 specimens.

Quantity of rice husk = 2.09 kg

Quantity of clay = 12.109 kg

OMC is 12.05

Therefore, quantity of water = 1.8 litre.

Table A.1 Replacement of clay with rice husk

Description	% variation	Rice husk (kg)	Clay (kg)	Water (litre)
Replacement of clay with RH	2%	0.299	14.651	1.94
	5%	0.74	14.202	1.89
	8%	1.196	13.754	1.83
	11%	1.64	13.30	1.8
	14%	2.09	12.109	1.8

A.2.3 Mix design for clay block partially replaced with red mud

Quantity for 5% of red mud replaced with clay, for 9 specimens.

Quantity of red mud = 0.744 kg

Quantity of clay = 14.20 kg

OMC is 13.09

Therefore, quantity of water = 1.95 litre.

Quantity for 15% of red mud replaced with clay, for 9 specimens.

Quantity of red mud = 2.24 kg

Quantity of clay = 12.70 kg

OMC is 13.2

Therefore, quantity of water = 1.97 litre.

Quantity for 25% of red mud replaced with clay, for 9 specimens.

Quantity of red mud = 3.73 kg

Quantity of clay = 11.21 kg

OMC is 13.68

Therefore, quantity of water = 2.04 litre.

Quantity for 35% of red mud replaced with clay, for 9 specimens.

Quantity of red mud = 5.23 kg

Quantity of clay = 9.71 kg

OMC is 13.74

Therefore, quantity of water = 2.05 litre.

Quantity for 45% of red mud replaced with clay, for 9 specimens.

Quantity of red mud = 6.72 kg

Quantity of clay = 8.22 kg

OMC is 13.79

Therefore, quantity of water = 2.05 litre.

Table A.2 Replacement of clay with red mud

Description	% variation	Red mud (kg)	Clay (kg)	Water (litre)
Replacement of clay with RM	5%	0.744	14.20	1.95
	15%	2.24	12.70	1.97
	25%	3.73	11.21	2.04
	35%	5.23	9.71	2.05
	45%	6.72	8.22	2.05

A.2.4 Mix design for clay block partially replaced with water hyacinth

Quantity for 1% of water hyacinth replaced with clay, for 9 specimens.

Quantity of water hyacinth = 0.149kg

Quantity of clay = 14.801 kg

OMC is 12.98

Therefore, quantity of water = 1.94 litre.

Quantity for 3% of water hyacinth replaced with clay, for 9 specimens.

Quantity of water hyacinth = 0.44 kg

Quantity of clay = 14.50 kg

OMC is 12.98

Therefore, quantity of water = 1.94 litre.

Quantity for 5% of water hyacinth replaced with clay, for 9 specimens.

Quantity of water hyacinth = 0.74 kg

Quantity of clay = 14.20 kg

OMC is 12.98

Therefore, quantity of water = 1.94 litre.

Quantity for 7% of water hyacinth replaced with clay, for 9 specimens.

Quantity of water hyacinth = 1.04 kg

Quantity of clay = 13.90 kg

OMC is 13.01

Therefore, quantity of water = 1.95 litre.

Quantity for 10% of water hyacinth replaced with clay, for 9 specimens.

Quantity of water hyacinth = 1.49 kg

Quantity of clay = 13.45 kg

OMC is 13.08

Therefore, quantity of water = 1.95 litre.

Table A.3 Replacement of clay with water hyacinth

Description	% variation	Water hyacinth (kg)	Clay (kg)	Water (litre)
Replacement of clay with WH	1%	0.149	14.801	1.94
	3%	0.44	14.50	1.94
	5%	0.74	14.20	1.94
	7%	1.04	13.90	1.95
	10%	1.49	13.45	1.95

A.2.5 Mix design for clay, rice husk, red mud & water hyacinth

Quantity for 11% of rice husk, 25% red mud and 5% water hyacinth replaced with clay, for 9 specimens.

Quantity of rice husk = 1.64 kg

Quantity of red mud = 3.73 kg

Quantity of water hyacinth = 0.74 kg

Quantity of clay = 8.82 kg

OMC is 13.24

Therefore, quantity of water = 1.97 litre